

Xcr1⁺ type 1 conventional dendritic cells are essential mediators for atherosclerosis progression

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This manuscript by Li, Lu et al., presents **important** findings on the role of cDC1 in atherosclerosis and their influence on the adaptive immune system. Using Xcr1Cre-Gfp Rosa26LSL-DTA ApoE^{-/-} mouse models, these data **convincingly** reveal an unexpected, non-redundant role of the XCL1-XCR1 axis in mediating cDC1 contributions to atherosclerosis.

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Abstract

Background

Atherosclerosis is characterized by lipid accumulation within plaques, leading to foam cell formation and an inflammatory response within the aortic lesions. Lipid disorders have been extensively investigated, however the cellular and molecular mechanisms that trigger the inflammatory response in atherosclerotic plaques remain far from being fully understood. Xcr1⁺ cDC1 cells are newly identified antigen-presenting cells in activating immune cells. However, the role of cDC1 cells in the development of atherosclerosis remains highly controversial.

Results

We first confirmed the presence of cDC1 within human atherosclerotic plaques and discovered a significant association between the increasing cDC1 numbers and

atherosclerosis progression in mice. Subsequently, we established $Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}$ mice, a novel and complex genetic model, in which cDC1 was constitutively depleted *in vivo* during atherosclerosis development. Intriguingly, we observed a notable reduction in atherosclerotic lesions in hyperlipidemic mice, alongside suppressed T cell activation of both $CD4^{+}$ and $CD8^{+}$ subsets in the aortic plaques. Notably, aortic macrophages and serum lipid levels were not significantly changed in the cDC1-depleted mice. Single-cell RNA sequencing revealed heterogeneity of $Xcr1^{+}$ cDC1 cells across the aorta and lymphoid organs under hyperlipidemic conditions. As $Xcr1$ is the sole receptor for $Xcl1$, we next explored to target $Xcr1^{+}$ cDC1 cells via $Xcl1$ by establishing $Xcl1$ and $ApoE$ deficient mice. $Xcl1$ knockout mice exhibited decreased atherosclerotic plaque formation and reduced aortic cDC1 accumulation, indicating that $Xcl1$ contributes to cDC1-mediated atherosclerotic lesion development.

Conclusions

Our results reveal crucial roles of cDC1 in atherosclerosis progression and provide insights into the development of immunotherapies by targeting cDC1 through $Xcl1$.

Introduction

Atherosclerosis is a chronic inflammatory disease that affects the large and medium-sized arteries, and it is the leading cause of cardiovascular mortality and morbidity worldwide [1]. Atherosclerosis is a complex pathological process during which lipid-laden phagocytes such as macrophages, along with lymphocytes and dendritic cells, accumulate in the plaque [2]. Mounting evidence from animal models and human studies demonstrates involvement of both the innate and adaptive immunity in the development of atherosclerosis [3–5]. In healthy arteries, DCs are typically present in the subendothelial space and work in the front line of immune surveillance [6, 7], and small portion of resident vascular DCs are localized in the adventitia. Various studies have demonstrated that DCs accumulated in the advanced atherosclerotic lesions in both humans and animal models [8–11]. Recently, several phenotypically and functionally distinct vascular DCs have been identified in mouse and human atherosclerotic lesions [12]. $Xcr1^{+}$ type 1 conventional dendritic cells ($Xcr1^{+}$ cDC1) (also known as $CD8^{+}$ DC or $CD103^{+}$ DCs) has been recognized for its role in cross-presentation and activating naive $CD8^{+}$ T cells, and are critical for anti-viral and anti-tumor immune responses [13, 14]. Owing to the challenges associated with obtaining a model for specific cDC1 depletion *in vivo*, previous studies investigating the role of cDC1 cells in atherosclerosis using various knockout mice have produced conflicting results [15–19]. It is important to note that previous Cre lines developed to achieve specific Cre expression in cDC1s still led to leakiness in recombinase activity outside of cDC1s, as demonstrated in previous publications [20, 21]. Moreover, scant mechanistic understanding regarding the regulation of cDC1 activity during atherosclerosis has been reported. Therefore, developing a more specific mouse model for cDC1 is essential for elucidating its precise role in atherosclerosis. Here, we established a novel Cre line $Xcr1^{Cre-Gfp}$ mice and did not observe any leakiness of Cre activity, and further experiments revealed that specific depletion cDC1 attenuates atherosclerosis development by inhibiting both $CD4^{+}$ and $CD8^{+}$ T cell activation in hyperlipidemic $Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}$ mice.

The origin of phagocytes within the atherosclerotic plaques has been controversial. Studies have shown that the major contribution of macrophages in the aortic plaque is recruited from the circulation, while other studies have emphasized the significance of local proliferation [22, 23]. Tissue-resident dendritic cells inside the plaques have been well documented [24, 25], regardless of the fact that the contribution and origin of $Xcr1^{+}$ cDC1 have not been appropriately addressed. To understand the origin of $Xcr1^{+}$ cDC1 inside the plaque, we performed bone marrow

transplantation using $Xcr1^{Cre-Gfp}$ $Rosa26^{LSL-DTA}$ mice as donors and found that in $ApoE^{-/-}$ recipient mice placed on a high-fat diet (HFD), cDC1 cells were completely absent in the aortas. Therefore, our experiments have elucidated that cDC1 cells in the atherosclerotic plaques originate from the bone marrow. Further single cell RNA sequencing illustrated that notable differences in $Xcr1^{+}$ cDC1 cells are discovered among the aorta, spleen, and lymph nodes. *Ccr2*, *Sept3* and *Cldnd1* were highly expressed in $Xcr1^{+}$ cDC1 cells from aorta compared with that from spleen and lymph nodes.

We further aimed to ascertain whether cDC1 can be modulated by targeting the chemokine *Xcl1*, which is a ligand of *Xcr1*. Considering the vital role that $Xcr1^{+}$ cDC1 plays in the progression of atherosclerosis and the therapeutic potential of manipulating their functions via chemokines [26], comprehending the feasibility of targeting *Xcl1* holds considerable promise for developing novel therapeutic approaches to halt the advancement of atherosclerosis. We conducted further studies using $Xcl1^{-/-}$ $ApoE^{-/-}$ double knockout mice to assess the pathological grade of atherosclerotic lesions. Our data indicated that the knockout of *Xcl1* led to a mild yet significant reduction of cDC1 cells in the aorta. However, strikingly, when compared to the $ApoE^{-/-}$ control mice, the $Xcl1^{-/-}$ $ApoE^{-/-}$ mice exhibited a markedly decreased severity of the atherosclerotic lesions, to an extent comparable to that observed in cDC1-depleted $Xcr1^{Cre-Gfp}$ $Rosa26^{LSL-DTA}$ $ApoE^{-/-}$ mice.

Therefore, our study which employed genetic models to deplete $Xcr1^{+}$ cDC1 *in vivo* in hyperlipidemic mice, has unequivocally determined the essential role of this specific cell type in the development of atherosclerosis. Besides their well-established role in cross-presentation to $CD8^{+}$ T cells, recent studies have shown $Xcr1^{+}$ cDC1 cells also engage with $CD4^{+}$ T cells to augment $CD8^{+}$ T cell responses [20, 27]. Our data showing $Xcr1^{+}$ cDC1-dependent CD4 and CD8 T cell activation in atherosclerotic plaques is in line with the new paradigm for cDC1s as a platform for both CD4 and CD8 T cell responses in the novel context of atherosclerosis.

Most importantly, we have successfully demonstrated that targeting the chemokine *Xcl1*, the ligand of *Xcr1*, can effectively inhibit atherosclerosis. These novel findings hold the potential to forge a novel trajectory for therapeutic interventions within the field of atherosclerosis, presenting promising prospects for better understanding and treatment of this disorder.

Results

Progressive accumulation of $Xcr1^{+}$ cDC1 cells in advanced atherosclerotic lesions in human and mice

Dendritic cells are present in healthy arteries and accumulate within atherosclerotic lesions, participating in various pathogenic and protective mechanisms during atherogenesis [28, 29]. However, the intra-plaque dynamics of cDC1 and their contribution to atherosclerosis remain undetermined. Using the archived specimens of paraffin-embedded human atherosclerotic arteries from our previous study [30], we examined the $Xcr1^{+}$ dendritic cells in human atherosclerotic plaques. As shown in **Figure 1A** and **1B**, $Xcr1^{+}$ dendritic cells were significantly more abundant in the plaque area of the human femoral artery compared to that in the plaque-free area. Similarly, the immunofluorescence staining results using specific antibodies for *Xcr1* and *CD11c* revealed that $Xcr1^{+}$ *CD11c^{+}* dendritic cells accumulated in the plaques of the aortic root in $ApoE^{-/-}$ mice that were placed on a HFD for 16 weeks (**Figure 1C** and **1D**). Sample preparations entailed sectioning and appropriate fixation of the aortic root tissues, which were described in greater detail in the Methods and Materials section. To further elucidate the dynamics of $Xcr1^{+}$ cDC1 cells within the plaque during the progression of atherosclerosis, we carried out immunohistochemical analyses of these cells in the lesioned aortic root (**Figure 1E**).

The area positive for Xcr1 expression was quantified at 8-week on a normal diet (ND), 8-week on HFD, 12-week on HFD and 16-week on HFD, respectively. Quantitative analysis was conducted using image processing software, revealing a continuous accumulation of Xcr1⁺ cDC1 cells in the lesioned area as the disease progressed (**Figure 1F**). Therefore, our experiments provided data indicating the enrichment of Xcr1⁺ cDC1 cells in aortic lesions in both humans and mice, and demonstrated that the accumulation of Xcr1⁺ cDC1 cells was significantly greater during the progression of atherosclerosis.

Xcr1^{Cre-Gfp} is selectively expressed in cDC1

To understand the role of Xcr1⁺ cDC1 in regulating the development of atherosclerosis, we established the Xcr1^{Cre-Gfp} ApoE^{-/-} mice in which the Cre recombinase and eGFP was co-expressed under the control of endogenous Xcr1 promoter following start codon (**Figure S1**). This new Cre line differs from a previous design in which Cre recombinase was expressed via IRES linkage at the 3' UTR and leaked expression of Cre was frequently observed [20, 31]. To verify the fidelity of Cre expression under hyperlipidemic condition, we used hyperlipidemic ApoE^{-/-} mice as recipient mice and used bone marrow cells from Xcr1^{Cre-Gfp} Rosa26^{LSL-tdRFP} mice as donors for the bone marrow transplantation (**Figure 2A**). In Xcr1^{Cre-Gfp} Rosa26^{LSL-tdRFP} mice, the presence of Cre protein would turn on the expression of tdRFP as a reporter for the activity of Cre recombinase [32]. Remarkably, we found that the expression of tdRFP was exclusively present in Xcr1⁺ cDC1 when mice were fed a HFD for 16 weeks in the bone marrow transplanted mice, and the Cre recombinase activity was not detectable in non-cDC1 cells (**Figure 2B** and **2C**).

To determine the specificity of GFP expression originating from Xcr1^{Cre-Gfp} genetic modification, we examined the GFP levels in three populations of DCs, namely pDC, cDC1, and cDC2 cells, from the spleen of Xcr1^{Cre-Gfp} ApoE^{-/-} mice that were fed a HFD for 16 weeks. As expected, we found that the GFP expression was stringently restricted to cDC1 cells (**Figure 2D** and **2E**). In parallel, we compared the GFP expression in Xcr1⁺ cDC1-depleted mice that were fed a HFD for 16 weeks. Such cDC1-depleted Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice were obtained by crossing Xcr1^{Cre-Gfp} ApoE^{-/-} mice with Rosa26^{LSL-DTA} ApoE^{-/-} mice. Strikingly, we observed a complete loss of GFP-positive cells and Xcr1⁺ cDC1 cells in Xcr1⁺ cDC1-depleted mice (**Figure 2D** and **2F**). We also examined DCs, including pDCs, cDC1, and cDC2, in the lymph nodes and spleens of Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} and ApoE^{-/-} mice maintained on a ND for 7 weeks. The flow cytometric analyses also confirmed the absence of Xcr1⁺ cDC1 population in Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice (**Figure S2**). Collectively, these data suggest that Cre recombinase activity was selectively confined to cDC1 cells, and no leakiness was observed in lineages other than the Xcr1⁺ cDC1 population. We also demonstrated that Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice constitute a desirable model for exploring the role of cDC1 in atherosclerosis.

The specific depletion of Xcr1⁺ cDC1 cells significantly alleviates atherosclerosis without influencing the lipid profile and the number of intra-plaque macrophages

To investigate the role of cDC1 cells in atherosclerosis, both Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} and the control ApoE^{-/-} mice were subjected to a HFD for 16 weeks. The ORO staining results demonstrated that the size of lesion in both descending aorta and aortic root was significantly smaller in Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice compared with ApoE^{-/-} control mice (12.82% vs 6.01% in descending aorta; 39.25% vs 33.57% in ORO-stained aortic root) (**Figure 3A-D**). Given that macrophages play a pivotal role in the development and progression of atherosclerotic lesions, particularly within the necrotic core [33, 34], we next investigated macrophage involvement in the aorta using IHC and flow cytometry. Interestingly, no significant differences were observed in the necrotic core area of the aortic root between Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice and ApoE^{-/-} control mice (**Figure 3E** and **3F**). Similarly, subsequent flow

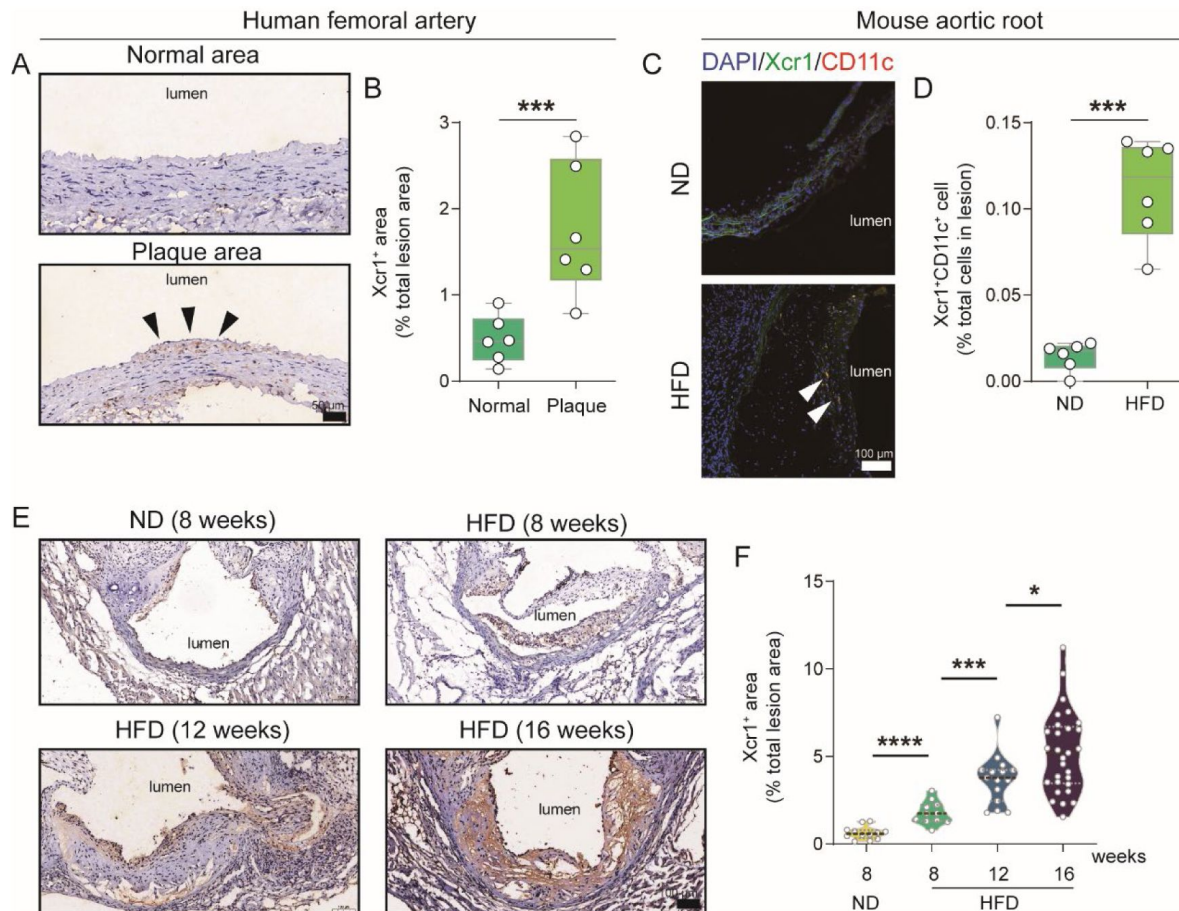


Figure 1.

Progressive accumulation of Xcr1⁺ cDC1 cells in human and mouse advanced atherosclerotic lesions

A, Representative images of immunohistochemical staining demonstrate Xcr1 expression in normal areas and plaque areas of the femoral artery from three cardiovascular disease patients. **B**, Quantitative analysis of the Xcr1-positive area in normal area and plaque regions. **C**, Representative immunofluorescence images depict DAPI (blue), Xcr1 (green) and CD11c (red) within the lesions of aortic root of ApoE^{-/-} mice fed with 16-week ND or HFD. **D**, Quantitative analysis of Xcr1 and CD11c double positive area in lesion area (n = 6). Scale bars, 100 μm. **E**, Representative images depicting immunohistochemical staining of Xcr1 in lesions of the aortic root of ApoE^{-/-} mice fed a HFD for varying durations. **F**, Quantitative analysis of the Xcr1-positive area in the lesion regions. (8-week HFD, n = 11; 12-week HFD, n = 15; 16-week HFD, n = 29). Data represent as mean ± SEM. * $P < 0.05$. *** $P < 0.001$. **** $P < 0.0001$.

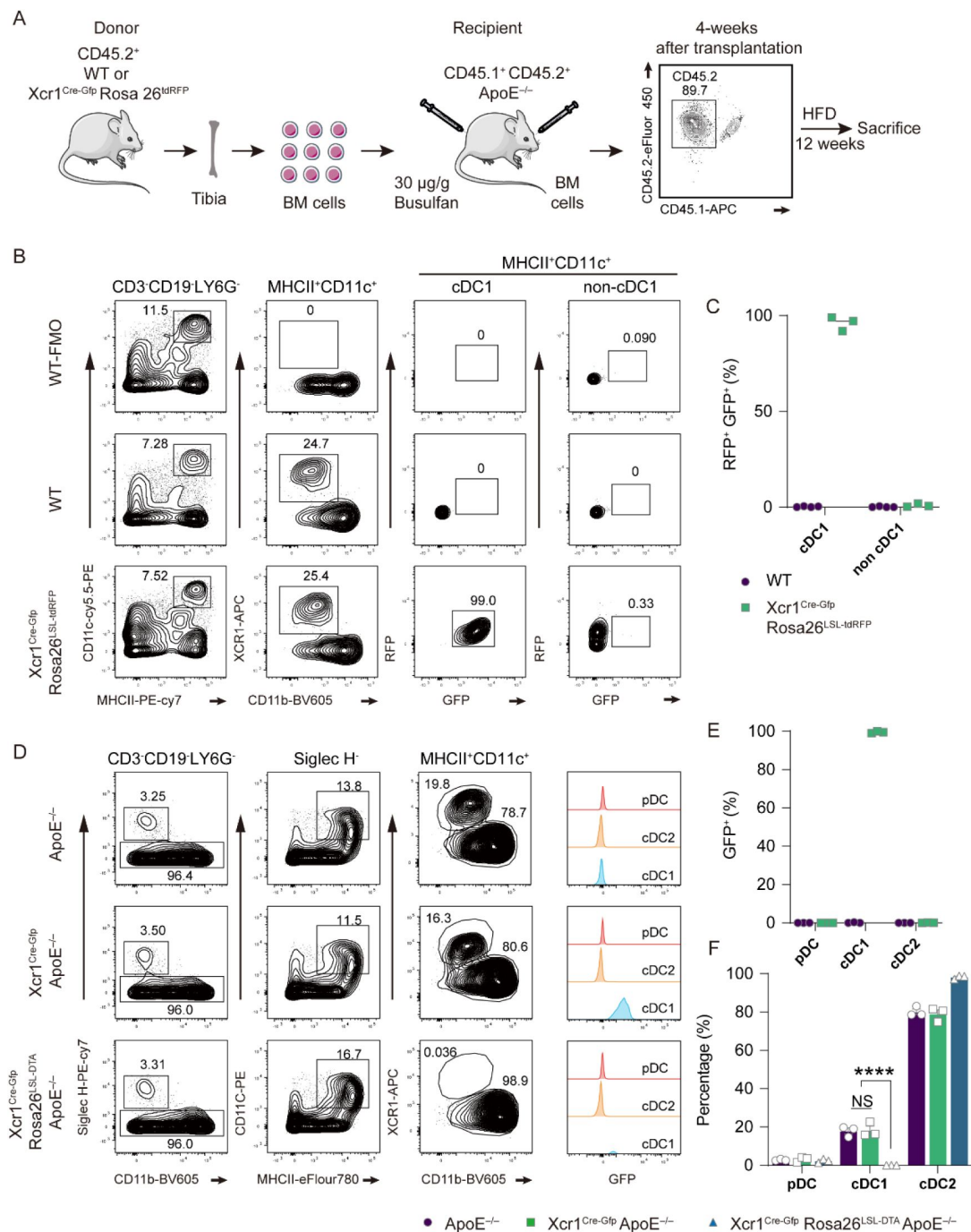


Figure 2.

Selective expression of *Xcr1*^{Cre-Gfp} in cDC1 cells.

A, Diagram illustrating the bone marrow transfer process. **B** and **C**, Representative flow cytometric analysis and corresponding quantification of GFP⁺RFP⁺ in cDC1 and non-cDC1 populations within the cDC1 and non-cDC1 populations of the spleen. These data were obtained from ApoE^{-/-} mice that had received bone marrow transplants from either WT or Xcr1^{Cre-Gfp} Rosa26^{LSL-RFP} donors and were maintained on a HFD for 16 weeks (WT, n = 4; Xcr1^{Cre-Gfp} Rosa26^{LSL-RFP}, n = 3). **D** and **E**, Representative flow cytometric analysis and quantification of the percentage of GFP⁺ cells among pDC, cDC1 and cDC2 cells populations in the spleen. **F**, Quantification of the percentage of pDC, cDC1 and cDC2 cells in spleen. (D-F, ApoE^{-/-}, Xcr1^{Cre-Gfp} ApoE^{-/-} and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed with a HFD for 16 weeks, n = 3). Data represent as mean ± SEM. **** P<0.0001; NS, non-significant.

cytometry analyses and IHC results revealed comparable proportions of F4/80⁺ CD11b⁺ macrophages in the aorta, as well as similar percentages of CD68-positive macrophages in the aortic lesions between the two groups of mice (**Figure 3G-J**). Furthermore, since dyslipidemia is a key characteristic of HFD-treated ApoE^{-/-} mice [35, 36], it is critical to assess body weight and lipid profiles in both groups. Our results indicated no significant differences in body weight or levels of total cholesterol (TC), triglycerides (TG), low-density lipoprotein (LDL), and high-density lipoprotein (HDL) between the two groups (**Figure S3A-B**). Additionally, ORO staining of the liver revealed no significant differences either (**Figure S3C-D**). Taken together, these results suggested that specific depletion of cDC1 attenuated atherogenesis without affecting lipid status or macrophagic accumulation in aortic lesions.

Specific depletion of cDC1 attenuates the development of atherosclerosis by modulating the activation of T cell in ApoE^{-/-} mice

T cells are among the critical drivers of the pathogenesis of atherosclerosis, and therefore, the triggers of T cell activation in aortic plaques are of great interest for controlling disease progression [2, 37, 38]. We initially analyzed the cDC1 cells through flow cytometry in both Xcr1⁺ cDC1 depleted mice and control mice. In the Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice that were fed a HFD for 16 weeks, we observed a total absence of Xcr1⁺ cDC1 in the aorta, lymph nodes, and spleen (**Figure 4A-C** and **Figure S4**). Subsequently, we assessed T cell phenotype in the two groups of mice. While neither the frequencies nor absolute counts of aortic CD4⁺ and CD8⁺ T cells differed significantly between two groups of mice (**Figure 4D-F**), CD69 frequency and CD44 MFI (Mean Fluorescence Intensity), the T cell activation markers, were significantly reduced in both CD4⁺ and CD8⁺ T cells from Xcr1⁺ cDC1 depleted mice compared to controls (**Figure 4G** and **H**). Interestingly, such decreased frequencies of CD69⁺ T cells were only observed in aortic lesions but not in lymphoid organs such as lymph nodes and the spleen in the Xcr1⁺ cDC1 depleted mice (**Figure S5A-B** and **S5D-E**). We further analyzed the absolute numbers of CD8⁺ and CD4⁺ T cells in the spleen and lymph nodes and found that the absolute numbers of CD8⁺ and CD4⁺ T cells were markedly reduced in the spleen but not in the lymph nodes in Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice compared to those in ApoE^{-/-} mice (**Figure S5C** and **S5F**). Absence of cDC1 also resulted in reduced frequencies of CD8⁺ T cells in both spleen and lymph nodes in hyperlipidemic mice. We also analyzed T cell populations in the lymph nodes and spleen in two groups of mice that were fed a ND and found that the frequencies of CD8⁺ T cells were significantly decreased in the lymph nodes of Xcr1⁺ cDC1 depleted mice (**Figure S6A-B**). More importantly, the frequencies of CD44^{high} CD62L^{low} activated memory cells within the CD8⁺ subsets in both the spleen and lymph nodes were significantly lower than those in the ApoE^{-/-} control mice (**Figure S6A-D**). However, in the CD4⁺ subsets, such frequencies of CD44^{high} CD62L^{low} activated cells were comparable in the lymph nodes but marginally lower in the spleen (**Figure S6A-D**). It is important to note that the decreased absolute number of CD8⁺ T cells in the spleen in Xcr1⁺ cDC1 depleted mice was only observed in HFD groups but not in ND groups (**Figure S5F** and **Figure S6F**). Our data indicate that in aortic plaques, Xcr1⁺ cDC1 cells are implicated in activating CD4⁺ and CD8⁺ T cells, and the loss of the cDC1 population led to a decreased absolute count of CD8⁺ T cells in the lymph nodes of HFD treated mice. Whereas in mice treated with a ND, the loss of the cDC1 population resulted in reduced frequencies of activated memory T cells but did not affect the absolute counts of T cells. Attenuated T cell activation might account for the mitigated atherosclerosis in Xcr1⁺ cDC1 depleted mice.

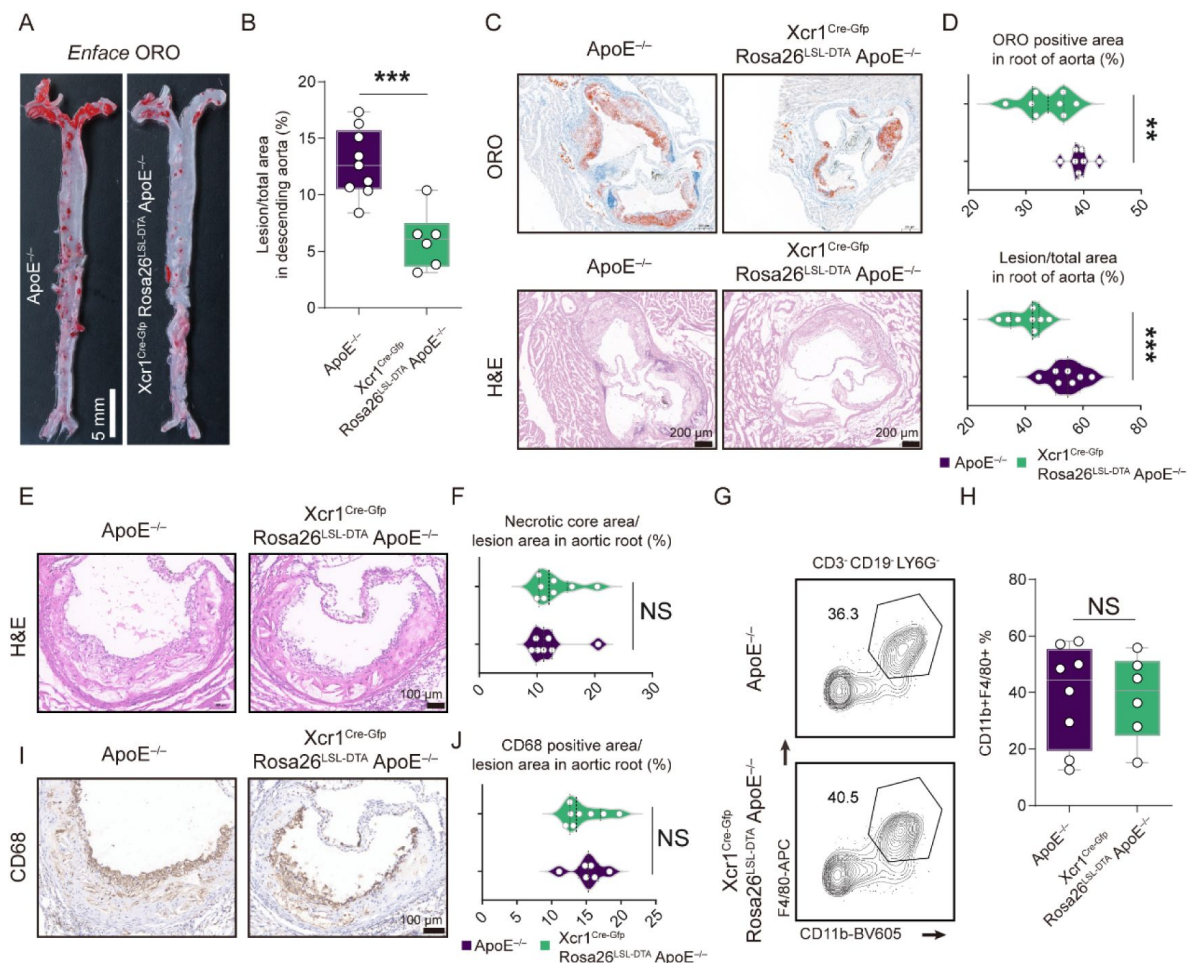


Figure 3.

Specific depletion of Xcr1⁺ cDC1 cells in ApoE^{-/-} mice reduces atherosclerosis progression with no effect on macrophages.

A, ORO staining of the descending aortas. Scale bar, 5 mm. **B**, Quantification of the lesion area in the descending aorta (ApoE^{-/-} mice, n = 9; Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, n = 6). **C**, ORO and H&E staining of the aortic roots of representative mice from each group. **D**, Quantification of lesion area in the aortic root. (ApoE^{-/-} mice, n = 7; Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, n = 8). Scale bar, 200 μ m. **E** and **F**, H&E staining and quantification of necrotic core area in the lesions of aortic root. (ApoE^{-/-} mice, n = 7; Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, n = 8). Scale bar, 200 μ m. **G** and **H**, Representative flow cytometric analysis and quantification of macrophages in the aorta. (ApoE^{-/-} mice, n = 8; Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, n = 6). **I** and **J**, Representative immunohistochemical staining images of CD68 and quantification of CD68 positive area in the lesions of the aortic root. (ApoE^{-/-} mice, n = 6; Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, n = 8). Scale bar, 100 μ m. ApoE^{-/-} and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice were fed on a 16-week HFD to develop atherosclerosis. Data represent as mean $\pm$ SEM. NS, non-significant. ** P<0.01. *** P<0.001.

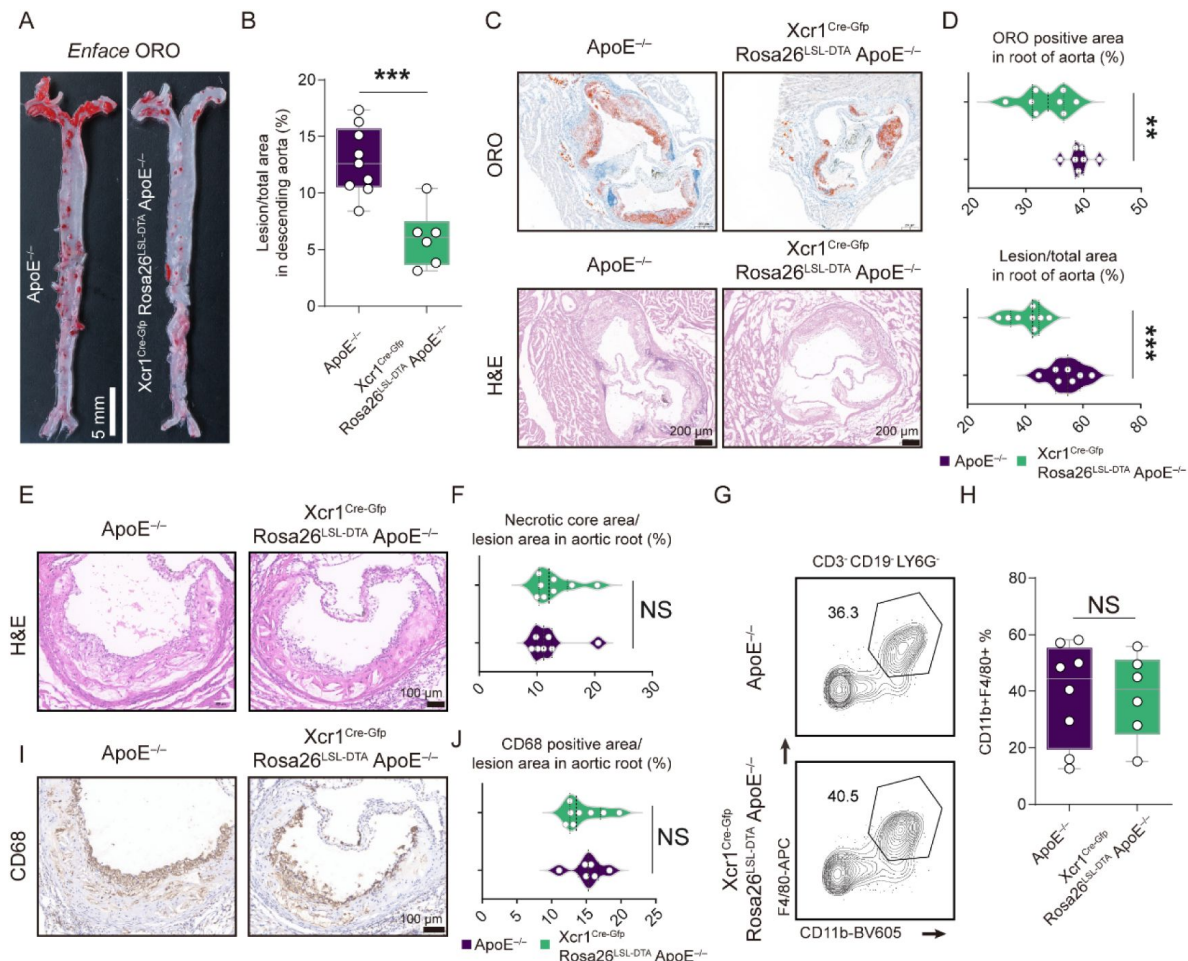


Figure 4.

Specific depletion of Xcr1⁺ cDC1 cells in ApoE^{-/-} mice decreases T cell activation within aorta.

A, Representative flow cytometric analysis cDC1 cells in aortas. **B** and **C**, Quantification of the frequencies and absolute counts of cDC1 cells in aorta (n = 4). **D**, Representative flow cytometric analysis T cells in aortas. **E** and **F**, Quantification of the frequencies and absolute counts of total T, CD4⁺ and CD8⁺ T cells in aortas. **G**, Quantification of the frequencies of CD4⁺CD69⁺ and CD8⁺CD69⁺ in aortas. **H**, Quantification of the MFI of CD44 in CD4⁺ and CD8⁺ T cells in aortas. (**D-G**, ApoE^{-/-} mice, n = 8, Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, n = 7, fed on a HFD for 16 weeks). Data represent as mean $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, NS, non-significant.

Characterization of aortic Xcr1⁺ cDC1 cells in atherosclerotic lesions

Although the majority of cDC1 cells derive from bone marrow, other mechanisms such as local proliferation of existing DCs can also contribute to the cDC1 population in inflammatory diseases [39, 40]. It remains unclear whether cDC1 cells in lymphoid organs such as spleen and lymph nodes represent a different status compared to cDC1 cells in atherosclerotic lesions. To elucidate the source and status of Xcr1⁺ cDC1 in atherosclerotic lesions, we performed bone marrow transplantation and single-cell RNA sequencing (sc-RNA seq) of cDC1 cells from aortic plaques and lymphoid organs. We conducted bone marrow transplantation using Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} or WT mice that carry the CD45.2 congenic markers as donors, and CD45.1⁺ CD45.2⁺ ApoE^{-/-} mice as recipients (Figure S7A). Our data showed that body weight, lipid profile, and hepatic lipid accumulation were comparable between two groups of bone marrow transplanted mice after being fed with a HFD for 16 weeks (Figure S7B-D). Interestingly, flow cytometric analysis of the aorta displayed almost complete depletion of cDC1 cells in ApoE^{-/-} recipient mice transplanted with Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} bone marrow mice (Figure S7E). Strikingly we did not find observe significant changes in pDCs and CD64⁺F4/80⁺ aortic macrophages (Figure S7E-F). Therefore, our bone marrow transplantation experiments further confirmed fidelity of the Cre recombinase expression in cDC1 cells in mice.

Next, we carried out sc-RNA seq to characterize Xcr1⁺ cDC1 cells from multiple organs of hyperlipidemic mice. Due to the scarcity of Xcr1⁺ cDC1 cells that can be sorted by FACS and processed for sc-RNA seq, we pooled FACS sorted Xcr1⁺ cDC1 cells from three organs of hyperlipidemic ApoE-deficient mice fed a HFD for 20 weeks, including the aorta, spleen, and lymph nodes, and Xcr1⁺ cDC1 cells from aorta and splenocytes of ApoE-deficient mice fed a ND. Libraries for sc-RNA seq were prepared using barcode oligos conjugated to anti-CD45 antibodies for surface staining so that samples from each type of organ could be compared in sequential analyses. Therefore, we had a total of five samples, and our focus was to understand the differences between Xcr1⁺ cDC1 cells from three different organs under hyperlipidemic conditions. The cells were sorted by gating Xcr1-positive cells out of the MHCII⁺CD11c⁺ conventional dendritic cells (Figure S8). Firstly, we obtained the UMAP plot of 10 clusters of cDC1 cells using the Seurat 4.0 package (Figure 5A). In this analysis, 3830 quality-controlled Xcr1⁺ cDC1 cells were from three types of organs in hyperlipidemic mice, and an additional 2081 cells were from cDC1 cells sorted from the spleen and aorta of ApoE deficient mice fed on a ND. The clusters were differentiated according to top 10 differentially expressed genes (Supplementary Table 1), and the heatmap shows marker genes and heterogeneity among the 10 clusters of cDC1 cells (Figure 5B). The purpose of this experiment was to determine whether cDC1 cells from different organs possess unique features and whether cDC1 cells from aortic lesions can be characterized by the expression of marker genes. In hyperlipidemic mice, cDC1 cells from the spleen and lymph nodes differed quite dramatically in Cluster 2 (*Stk17b* and *Gpr171* highly expressing cells), where 28.9% of cDC1 cells from the spleen versus 1.0% of cDC1 cells from the lymph nodes were located. Conversely, 3.15% of cDC1 cells from the spleen versus 32.4% of cDC1 cells from the lymph nodes fell in Cluster 3 (*Ccl5* and *Eptsi1* highly expressing cells), and 4.6% of cDC1 cells from the spleen versus 25.9% of cDC1 cells from the lymph nodes fell in Cluster 4 (*Slfn5* and *Ifi44* highly expressing cells). Despite the fact that we had only transcriptomic data for 68 Xcr1⁺ cDC1 cells from aortic lesions, we were surprised to find that such aortic cDC1 mainly fell into Cluster 1, 4 and 5, comprising 80.8% of the cells analyzed (Figure 5C). In Cluster 1 (*Hspa1b* and *Dnajb1* highly expressing cells), where aortic cDC1 is dominantly enriched, the cDC1 cells sorted from lymph nodes were essentially absent, while in Cluster 4, aortic cDC1 cells were highly enriched and splenic cDC1 was obviously lower in percentage (Figure 5C). Cluster 7 (*Fabp5* and *S100a4* highly expressing cells) is exclusively comprised of cDC1 cells sourced from lymph nodes, with a striking absence of cDC1 cells from the spleen and aorta (Figure 5C). To better visualize

the enrichment of aortic $Xcr1^+$ cDC1 cells distributed in a few clusters, we separately compared the three samples from the spleen, lymph nodes, and aorta of hyperlipidemic mice using density plots. Interestingly, the $Xcr1^+$ cDC1 cells from the spleen and lymph nodes displayed an obvious difference in their distribution, and aortic $Xcr1^+$ cDC1 cells appeared more enriched in a few clusters (**Figure 5D-E**). Therefore, our data suggest that $Xcr1^+$ cDC1 cells from the spleen and lymph nodes can differ markedly in heterogeneity, and aortic $Xcr1^+$ cDC1 cells are enriched in a few clusters that express marker genes, *Ccr2*, *Sept3* and *Cldnd1* (**Supplementary Table 2**). Using the scRNA seq data of $Xcr1^+$ cDC1 sorted from hyperlipidemic mice, we annotated the 10 populations as shown in **Figure S9A**, following the methodology from a previous study [41]. *Ccr7*⁺ mature cDC1s (Cluster 3, 7 and 9) and *Ccr7*-immature cDC1s (remaining clusters) were identified across cDC1 cells sorted from aorta, spleen and lymph nodes (**Figure S9B**). Further stratification based on marker genes reveals that Cluster 10 is the pre-cDC1, with high expression level of *CD62L* (*Sell*) and low expression level of *CD8a* (**Figure S9C**). Cluster 6 and 8 are the proliferating cDC1s, which express high level of cell cycling genes *Stmn1* and *Top2a* (**Figure S9D**). Cluster 1 and 4 are early immature cDC1s, and cluster 2 and 5 are late immature cDC1s, according to the expression pattern of *Itgae*, *Nr4a2* (**Figure S9E**). Cluster 9 cells are early mature cDC1s, with elevated expression of *Cxcl9* and *Cxcl10* (**Figure S9F**). Cluster 3 and 7 as late mature cDC1s, characterized by the expression of *Cd63* and *Fscn1* (**Figure S9G**). As shown in **Figure 5C** and **Figure S9**, the 10 populations displayed a major difference of aortic cDC1 cells that lack in pre-cDC1s (cluster 10) and mature cells (cluster 3, 7 and 9). Interestingly, in hyperlipidemic mice splenic cDC1 possess only Cluster 3 as the late mature cells while the lymph node cDC1 cells have two late mature populations namely Cluster 3 and Cluster 7. In further analysis, we also compared splenic cDC1 cells from HFD mice to those from ND mice. As shown in **Figure S10**, HFD appears to impact early immature cDC1-1 cells (Cluster 1) and increases the abundance of late immature cDC1 cells (Cluster 2 and 5), regardless of the fact that all 10 populations are present in two origins of samples. We also found that *Tnfrsf3* and *Serinc3* are among the most upregulated genes, while *Apol7c* and *Tifab* are downregulated in splenic cDC1 cells sorted from HFD mice.

Together, these data implied that $Xcr1^+$ cDC1 in the aorta originate from bone marrow, and $Xcr1^+$ cDC1 cells exhibit strong heterogeneity among aorta, spleen and lymph nodes.

Knockout of *Xcl1* attenuates the development of atherosclerosis in $ApoE^{-/-}$ mice

Xcl1 is the major chemokine involved in cDC1 recruitment [42, 43], and plays critical role in immune response with diverse physical and pathological implication through its interaction with the sole receptor, *Xcr1* [44, 45]. We found the presence of *Xcl1* in the lesion area in both human and mouse atherosclerotic lesions (data not shown). To elucidate the causal relationship between *Xcl1* and $Xcr1^+$ cDC1 in the development of atherosclerosis, and more importantly to experimentally verify *Xcl1* as a potential therapeutic target for atherosclerosis treatment, we established the *Xcl1* and *ApoE* double knockout mice. After being fed with a HFD for 19 weeks, both $Xcl1^{-/-}$ $ApoE^{-/-}$ and $ApoE^{-/-}$ mice were sacrificed to assess the severity of atherosclerosis. As shown in **Figure 6A-D**, the size of lesion at the descending aorta and aortic root was significantly smaller in $Xcl1^{-/-}$ $ApoE^{-/-}$ compared to that in $ApoE^{-/-}$ mice. Importantly, ORO staining of the liver, body weight and blood lipid profiles revealed no significant differences between the two groups (**Figure 6E-H**). Moreover, the IHC experiment result show the percentage of CD68 positive macrophages in aortic lesion was similar between two groups (**Figure 6I-J**).

Previous study identified that *Xcl1* serves as a potent chemokine, selectively acting on cDC1 cells through its exclusive receptor, *Xcr1* [43]. Consequently, we examined the DCs and the activation of T cells in different organs, including aorta, lymph node and spleen. The flow cytometric results illustrated that both frequencies and absolute counts of $Xcr1^+$ cDC1 cells in the aorta were significantly reduced, but cDCs and cDC2 cells from $Xcl1^{-/-}$ $ApoE^{-/-}$ were comparable with that

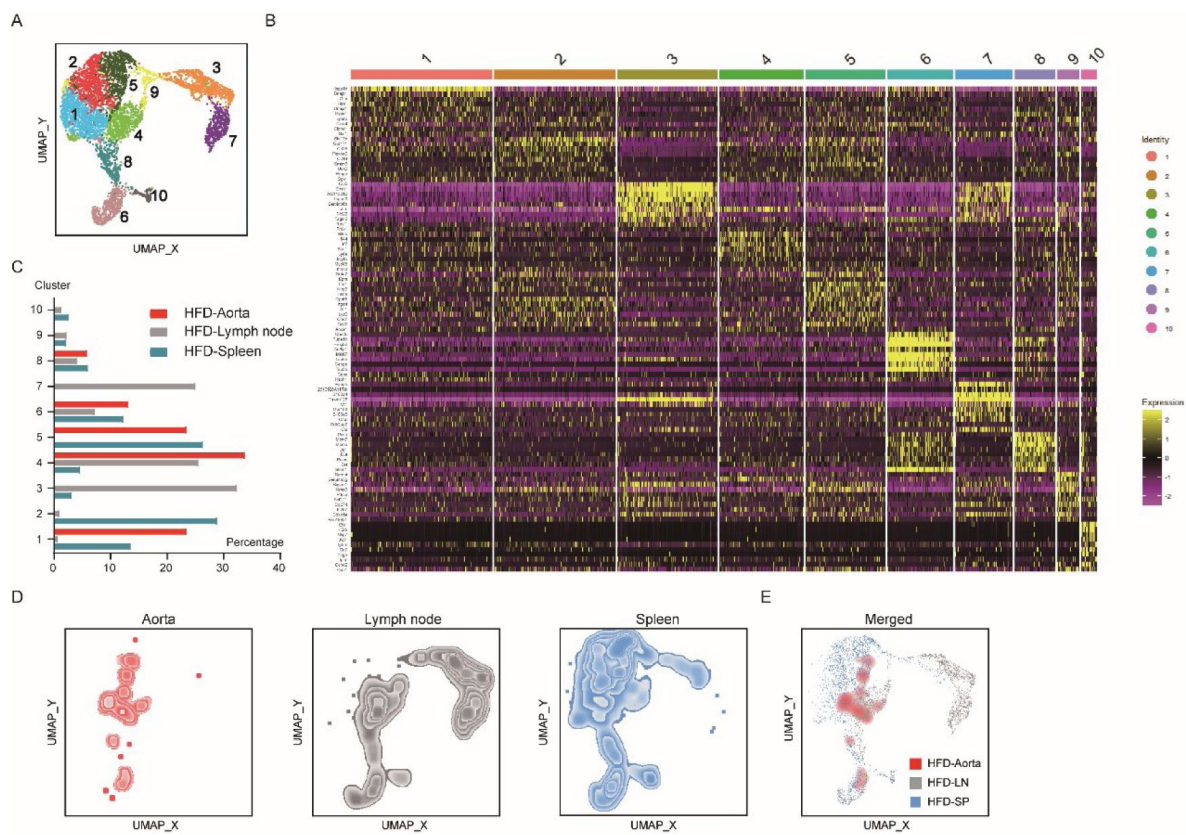


Figure 5.

Sc-RNA sequencing analysis of the aorta, lymph nodes and spleen in ApoE^{-/-} mice fed with a 20-week HFD.

A, UMAP plot delineates 10 annotated cell types of Xcr1⁺ cDC1 cells from aorta, lymph node and spleen in ApoE^{-/-} mice maintained on a HFD or ND for 20 weeks. **B**, Heatmap statistic map displays the top ten up-regulated genes across various clusters. **C**, The proportion of 10 clusters among groups. **D** and **E**, UMAP plot representing Xcr1⁺ cDC1 cells from aorta, lymph nodes, spleen, and a merged plot in ApoE^{-/-} mice maintained on a HFD for 20 weeks.

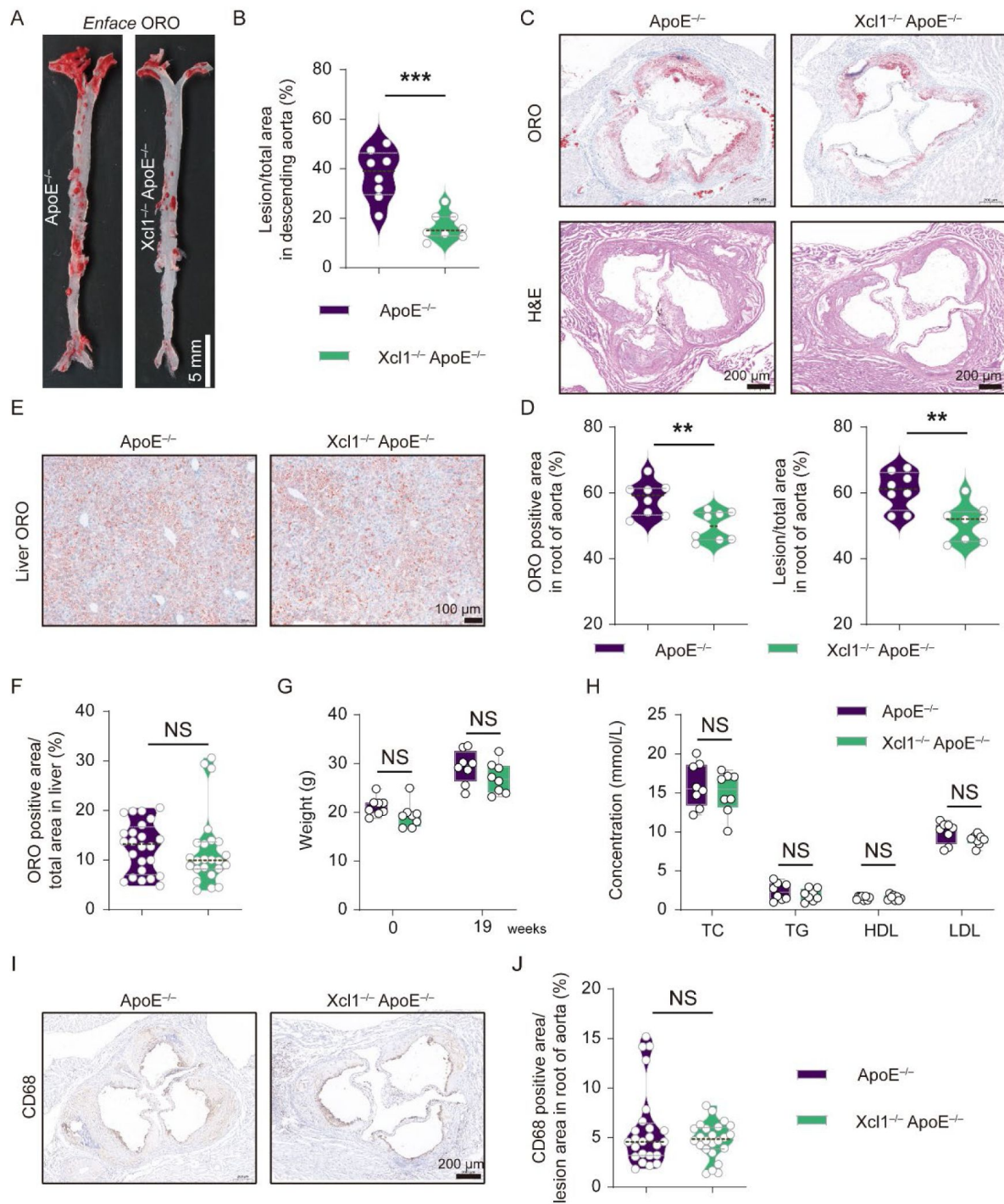


Figure 6.

Xcl1 deficiency reduces atherosclerotic lesion formation in ApoE^{-/-} mice.

A, ORO staining of the descending aortas. Scale bar, 5 mm. **B**, Quantification of the lesion area in aortas. **C**, ORO and H&E staining of aortic roots. Scale bar, 200 μ m. **D**, Quantification of the lesion area in aortic roots. **E** and **F**, Representative liver images and quantification of ORO positive area in livers (n = 24 regions from 8 mice, 3 regions per mouse). Scale bar, 100 μ m. **G**, The concentrations of TC, TG, LDL and HDL in the serum. **H**, Body weight of mice before and after feeding with 19-week HFD. **I**, Representative immunohistochemical staining of CD68 in aortic roots. **J**, Quantification of CD68 positive area in the lesions of aortic roots (n = 24 lesions from 8 mice, 3 lesions per mouse). Scale bar, 100 μ m. ApoE^{-/-} mice and Xcl1^{-/-} ApoE^{-/-} mice fed a HFD for 19 weeks (n = 8 per group). Data represent as mean $\pm$ SEM. ** P<0.01, *** P<0.001, NS, non-significant.

from ApoE^{-/-} (Figure 7A-C). Moreover, in both lymph node and spleen, the absolute numbers of pDC, cDC1 and cDC2 from Xcl1^{-/-} ApoE^{-/-} were comparable with that from ApoE^{-/-} (Figure S11). Crucially, CD69⁺ frequency and CD44 MFI remained comparable in both aortic CD4⁺ and CD8⁺ T cells between two groups (Figure 7D-F). However, aortic CD8⁺ T cells exhibited reduced frequency and absolute count, while CD4⁺ T cells showed increased frequency but unchanged counts in Xcl1^{-/-} ApoE^{-/-} mouse versus controls (Figure 7G and H). Furthermore, the absolute numbers of CD4⁺, CD8⁺, CD4⁺ CD69⁺ and CD8⁺ CD69⁺ T cells in both lymph node and spleen from Xcl1^{-/-} ApoE^{-/-} mouse were comparable with that from ApoE^{-/-} mouse (Figure S12).

Collectively, these data suggest that Xcl1 promotes atherosclerosis by recruiting Xcr1⁺ cDC1 cells, and facilitating CD8⁺ T cell accumulation in lesions. More importantly, *in vivo* data demonstrated the promising potential of targeting Xcl1 to reduce the inflammatory response in atherosclerotic lesions. Our work paves the way to discover novel solutions for the treatment of atherosclerosis by targeting Xcl1 and Xcr1⁺ cDC1 cells.

Discussion

Atherosclerosis is a complex disorder, and understanding the role of the immune system, particularly dendritic cells (DCs), is of paramount importance [46, 47]. The marked diversity of DC subsets and their distinct, and at times contradictory functions pose challenges in clarifying the role of each specific type of DC in the context of atherosclerosis [48–54]. Moreover, the restricted characterization of each subset of DCs in atherosclerosis, especially when compared to that in well-studied organs such as the skin and lymph nodes [13], has been severely hindered by the limited application of appropriate genetic models and *in vivo* data. Equally significant is that dissecting the functions of each subset of DCs *in vivo* relies on the development of genetic tools validated for lineage specificity. Recently, Xcr1⁺ cDC1 has been extensively investigated using the Cre line under the promoter of the Xcr1, despite well-documented expression leakiness [20, 31]. Compelling evidence from tumor and viral infection models [14, 42, 55–57], as well as metabolic disorders like non-alcoholic steatohepatitis [58], has demonstrated the critical role of Xcr1⁺ cDC1 cells in diseases. Nevertheless, the precise role of Xcr1⁺ cDC1 cells in atherosclerosis remains undetermined.

The precise characterization of Xcr1⁺ cDC1 and its functions within the atherosclerotic milieu could provide valuable insights into the immunopathogenesis of the disease. By delineating the specific contributions of Xcr1⁺ cDC1, we may identify novel therapeutic targets and develop more targeted and effective strategies for the prevention and treatment of atherosclerosis. Therefore, investigation to understand the roles of Xcr1⁺ cDC1 in the context of atherosclerosis and to identify therapeutic targets is highly warranted. However, at least two major obstacles impede the *in vivo* study of Xcr1⁺ cDC1 in atherosclerotic mice. First, Xcr1⁺ cDC1 cells are extremely scarce in number, and it is quite challenging to obtain highly purified Xcr1⁺ cDC1 cells from aortic plaque in hyperlipidemic mice. Second, there are very few appropriate genetic model systems that permit functional analyses of Xcr1⁺ cDC1 *in vivo*. Earlier studies on the role of Xcr1⁺ cDC1 cells in atherosclerosis have yielded conflicting results, largely due to the use of different mouse models. For instance, the deletion of Batf3, Irf8, Dnrg1, or Flt3 decreased the cDC1 cells. Nevertheless, Batf3 deficiency had no impact on atherosclerosis in hyperlipidemic mice. In contrast, Irf8 or Dnrg1 deficiency reduced the development of atherosclerosis, while Flt3 deficiency exacerbated the condition [15–19]. The recently developed Xcr1-Cre was found to have leakiness of recombinase activity in other immune cells [20]. Here, we optimized the design for the construction of the Cre line using the endogenous promoter of Xcr1 and validated its fidelity under hyperlipidemic conditions. We crossed the novel knock-in line termed Xcr1^{Cre-Gfp} with Rosa26^{LSL-tdRFP} Cre reporter mice and used the resulting F1 hybrids as bone marrow donors. The bone marrow cells were then transferred into ApoE^{-/-} mice, which were subsequently fed on a HFD. We

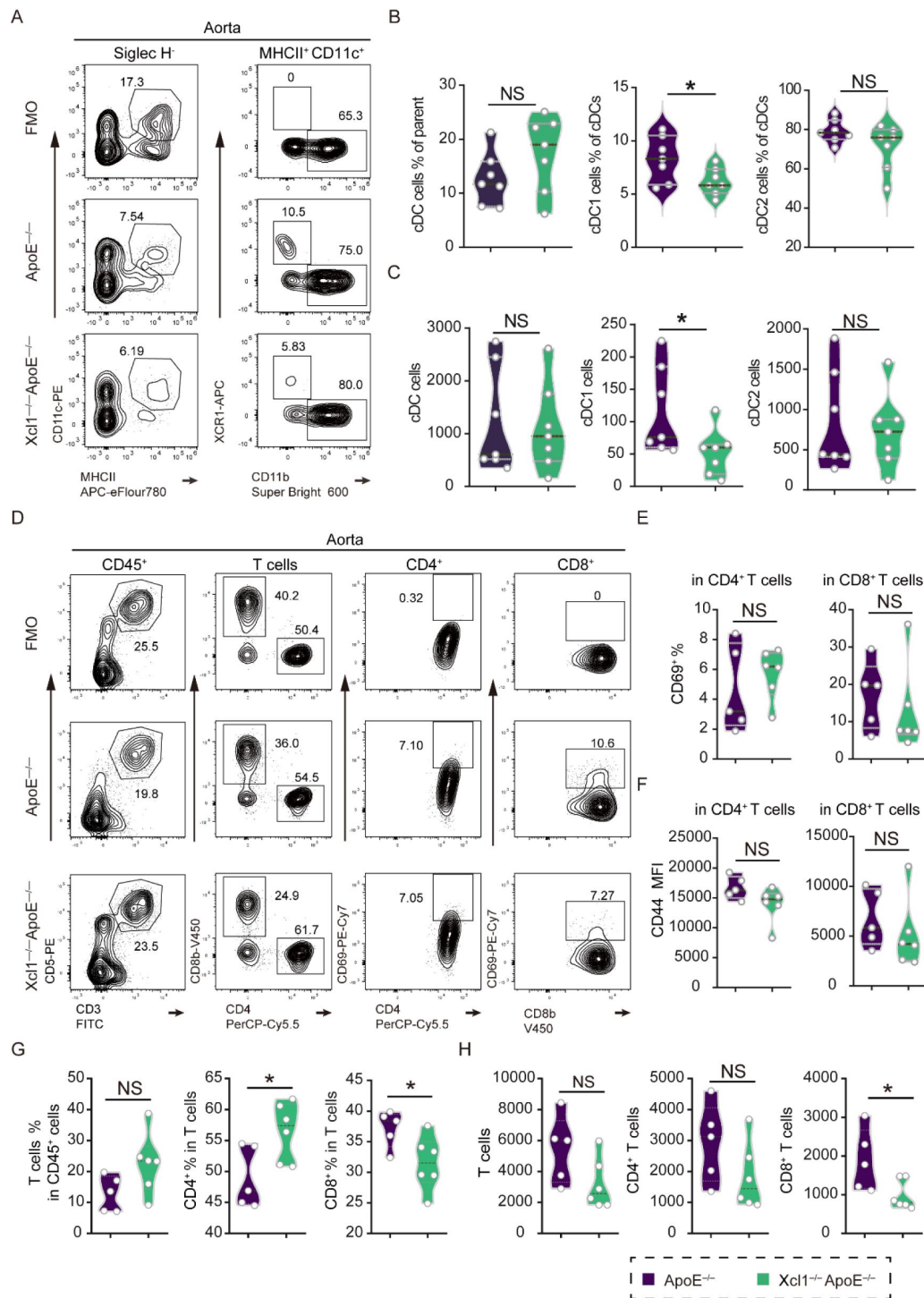


Figure 7.

Xcl1 deficiency reduces CD8⁺ T cells in the aortas from ApoE^{-/-} mice.

A, Representative flow cytometric analysis of cDC, cDC1 and cDC2 cells in aortas. **B** and **C**, Quantification of the frequencies and absolute counts of cDC, cDC1 and cDC2 cells in aortas (n = 7). **D**, Representative flow cytometric analysis of T cell subsets in aortas, including CD3e⁺ CD5⁺, CD4⁺, CD4⁺ CD69⁺, CD8⁺ and CD8⁺ CD69⁺ T cells. **E**, Quantification of the frequencies of CD4⁺CD69⁺ and CD8⁺CD69⁺ in aortas. **F**, Quantification of the MFI of CD44 in CD4⁺ and CD8⁺ T cells in aortas. **G** and **H**, Quantification of the frequencies and absolute counts of CD3e⁺ CD5⁺, CD4⁺ and CD8⁺ cells in aortas. (D-H, ApoE^{-/-} mice (n = 5) and Xcl1^{-/-} ApoE^{-/-} mice (n = 6) fed with a 16-week HFD). Data represent as mean ± SEM. * P<0.05, NS, non-significant.

monitored the Cre recombinase activity, which was reflected by the tdRFP expression of this novel mouse line. Under hyperlipidemic conditions, we did not observe any leakiness of Cre activity, as tdRFP was strictly restricted to cDC1 cells. To the best of our knowledge, this novel Cre line is the first of its kind. We further crossed the $Xcr1^{Cre-Gfp}$ and $Rosa26^{LSL-DTA}$ mice with $ApoE^{-/-}$ mice respectively. Subsequently, by further crossing, we obtained $Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}$ mice to create a complex genetic model, in which $Xcr1^{+}$ cDC1 is depleted *in vivo* in the context of atherosclerosis. Very interestingly, in the $Xcr1^{+}$ cDC1 depleted mice, atherosclerosis was significantly ameliorated. Coupled with the data we obtained from murine and human samples, where $Xcr1^{+}$ cDC1 cells exist in the aortic plaque and their accumulation is correlated with disease progression, we conclude that $Xcr1^{+}$ cDC1 is essential for the development of atherosclerosis.

Despite the fact that $Xcr1^{+}$ cDC1 cells are extremely rare in number, we were inclined to understand if there were any differences among $Xcr1^{+}$ cDC1 cells from different organs. Firstly, we determined that the $Xcr1^{+}$ cDC1 in the aortic plaque is replenished by circulating precursors using the bone marrow transfer model. Then we compared, under hyperlipidemic conditions, the $Xcr1^{+}$ cDC1 cells sorted from the spleen and lymph nodes, supplemented by a small number of but well-purified $Xcr1^{+}$ cDC1 cells from aortic lesions. To our surprise, the clusters of $Xcr1^{+}$ cDC1 displayed obvious distinctions among different organs. When the $Xcr1^{+}$ cDC1 cells from aortic lesions were projected to the clusters, we found that the major subsets present in the lymph nodes, characterized by the expression of markers *Ccl5* and *Fabp5*, and in the spleen, marked by *Stk17b* and *Gpr171*, were basically absent in the aorta. Meanwhile, compared with spleen and lymph nodes, the aortic $Xcr1^{+}$ cDC1 cells are highly expressed *Ccr2*, which is an extremely important chemokine receptor dominating monocyte homing [59–61].

This pronounced tissue-specific compartmentalization of $Xcr1^{+}$ cDC1 subsets may related to multiple mechanisms including developmental imprinting that instructs precursor differentiation into transcriptionally distinct subpopulations [62], and microenvironmental filtering through organ-specific chemokine axes (e.g., CCL2/CCR2 in spleen) selectively recruits receptor-matched subsets [63, 64]. This spatial specialization optimizes pathogen surveillance for local immunological challenges. Based on the maturation analysis of the cDC1 scRNA seq data [41], our findings suggest that the aortic cDC1 cells display a major difference from those of spleen and lymph nodes by lacking the mature clusters, whereas lymph node cDC1 cells contain an additional *Fabp5⁺ S100a4⁺* late mature Cluster. Our results also suggest that hyperlipidemia contributes to alteration in early immature cDC1 and in the abundance of late immature cDC1 cells, which was associated with dramatic change in gene expression of *Tnfrsf3*, *Serinc3*, *Apol7c* and *Tifab*.

In other disease models, $Xcr1^{+}$ cDC1 cells are crucial for activating T cells [56, 65], and we confirmed that in hyperlipidemic mice, the depletion of cDC1 substantially attenuated T cell activation responses. It is important to note that $Xcr1^{+}$ cDC1 cells are known for well-established role in cross-presentation to CD8⁺ T cells, recent studies have shown $Xcr1^{+}$ cDC1 cells also engage with CD4⁺ T cells [20, 27]. Our data showing $Xcr1^{+}$ cDC1-dependent CD4 and CD8 T cell activation in atherosclerotic plaques is in line with the new paradigm for cDC1s as a platform for both CD4 and CD8 T cell responses in the novel context of atherosclerosis. Considering the multifaceted roles of macrophages in lipid metabolism, inflammation, and plaque dynamics during the development of atherosclerosis [66], we evaluated the presence of macrophages in atherosclerotic lesions using IHC and flow cytometry. Our results indicated comparable proportions of macrophages in the aorta between $Xcr1^{+}$ cDC1 depleted mice and $ApoE^{-/-}$ control mice. Additionally, our results also demonstrated that the loss of $Xcr1^{+}$ cDC1 cells did not affect lipid status.

Since we elucidated the critical role of $Xcr1^{+}$ cDC1 cells in atherosclerosis, our aim was to test the therapeutic potential of inhibiting this specific cell type during disease progression. As $Xcr1$ is the sole receptor for Xcl1, and Xcl1 selectively attracts $Xcr1^{+}$ cDC1 cells in both mice and humans [45, 67], we performed Xcl1 knockout on the background of $ApoE$ deficient mice and assessed

the impact on atherosclerosis development. We found that the Xcl1-Xcr1 axis plays a crucial role in the progression of atherosclerosis. Our results demonstrated that the loss of Xcl1 in hyperlipidemic ApoE^{-/-} mice significantly reduced atherosclerotic lesion formation, decreased the frequencies and absolute counts of Xcr1⁺ cDC1 cells and CD8⁺ T cells in the aorta, without affecting lipid status or the number of aortic macrophages. Notably, while complete ablation of Xcr1⁺ cDC1s impaired T cell activation, reduction of Xcr1⁺ cDC1 recruitment via Xcl1 deletion did not significantly compromise this process. This discrepancy may arise through compensatory mechanisms: alternative chemokine axes (e.g., CCL5/CCR5, CXCL9/CXCR3, BCL9/BCL9L) may partially rescue Xcr1⁺ cDC1 homing [13, 68, 69], while non-cDC1 antigen-presenting cells (e.g., monocytes, cDC2s) may sustain T cell activation [55, 70]. Furthermore, tissue-specific microenvironment factors could potentially modulate its role in other diseases. In summary, our findings identify Xcl1 as a potential therapeutic target for atherosclerosis therapy, though its cellular origins and regulation of lesional Xcr1⁺ cDC1 and T cells dynamics require further studies.

Materials and Methods

Animal experiments and patients

All mice were raised in a Specific Pathogen-Free facility and the experiments were conducted in accordance with the guidelines approved by the Animal Ethics Committee of Xinxiang Medical University (China) (Reference No. XYLL-2016S001). The sections of the human femoral artery were obtained in accordance with our previous study [30] and approved by the Ethics Committee of the Third Military Medical University (Project identification code SYXK-PLA-20170005). Rosa26^{LSL-RFP} knock-in mice were generously provided by Hervé Luche at University Clinics Ulm in Germany [32]. C57BL/6 and ApoE^{-/-} mice were purchased from Beijing Vital River Laboratory Animal Technology (China). Rosa26^{LSL-DTA} knock-in mice (Stock Number 009669) were purchased from the Jackson Laboratory (United States of America). Both HFD (contained 20% protein, 20% carbohydrate, 40% fat and 1.25% cholesterol, Cat: XT108C), and ND (contained 20% protein, 64% carbohydrate, 20% fat, Cat: AIN-93G) were purchased from Jiangsu Xietong Pharmaceutical Bioengineering Co., Ltd. 7-8 weeks old mice were used to establish atherosclerosis model through fed on a HFD for 8-20 weeks. On harvest, each sample was assigned an analytical code that was irrelevant to its genotype and processed and analyzed by a researcher blinded to its origin.

Generation of Xcr1^{Cre-Gfp} knock-in mice

The Xcr1^{Cre-Gfp} mouse line was generated by inserting a Cre-P2A-GFP-P2A cassette immediately upstream of the endogenous *Xcr1* translational start codon (ATG). This design enables co-expression of Cre recombinase and GFP under the control of the native *Xcr1* promoter. A 3.2-kb repair template was synthesized (GenScript, Nanjing, China), consisting of a 0.62-kb 5' homology arm (HA), a codon-optimized *Cre* sequence fused to a P2A peptide, an enhanced GFP reporter linked to a second P2A site, and a 0.67-kb 3' homology arm. For targeted integration, CRISPR/Cas9-mediated homology-directed repair (HDR) was performed by co-injecting Cas9 mRNA, a chimeric sgRNA (sgXcr1-KI), and the repair template into fertilized C57BL/6 zygotes at the one-cell stage, following established protocols [71, 72]. Founders (F0) harboring the correct knock-in allele were identified by PCR and Sanger sequencing, then used to establish heterozygous lines. The sgRNA sequence is provided in **Supplementary Table 3**.

Breeding strategy of Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}

To generate Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice, we performed the following crosses: Xcr1^{Cre-Gfp} mice were crossed with ApoE^{-/-} mice, and offspring heterozygous for both alleles were intercrossed to obtain homozygous Xcr1^{Cre-Gfp} ApoE^{-/-} mice. Rosa26^{LSL-DTA} mice were crossed with ApoE^{-/-} mice, and progeny heterozygous for both alleles were intercrossed to generate

homozygous $Rosa26^{LSL-DTA} ApoE^{-/-}$ mice. Finally, homozygous $Xcr1^{Cre-Gfp} ApoE^{-/-}$ mice were crossed with homozygous $Rosa26^{LSL-DTA} ApoE^{-/-}$ mice to produce $Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}$ offspring.

Generation of Xcl1 knockout mice

The Xcl1 knockout mouse line was generated using CRISPR/Cas9 genome editing through the microinjection of a mixture containing Cas9 mRNA and two sgRNAs into fertilized C57BL/6 zygotes at the one-cell stage, as previously described [73]. The genotypes were confirmed using PCR and Sanger sequencing. Subsequently, Xcl1 and ApoE double knockout mice were obtained by crossing the Xcl1^{-/-} mice with ApoE^{-/-} mice. The sequences of the sgRNA are reported in **Supplementary Table 3**.

Bone marrow transplantation

After one week of antibiotic treatment, the recipient CD45.1⁺ CD45.2⁺ ApoE^{-/-} mice were injected intraperitoneally with busulfan at a dosage of 30 µg/g every two days for one week. Subsequently, bone marrow from 8-week-old CD45.2⁺ Xcr1^{Cre-Gfp} Rosa26^{LSL-RFP}, CD45.2⁺ Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} or WT (CD45.2⁺ wild-type) donor mice was intravenously transferred to each recipient mouse, with two million cells administered per recipient. The efficiency of transplantation was assessed four-week post-transplantation, after which all mice were fed a HFD for 16 weeks. Finally, the mice were sacrificed for *en face* analysis of the descending aorta, hematoxylin and eosin (H&E) staining, Oil Red O (ORO) staining of the aortic root, and flow cytometry analysis, as previously described [74].

Immunofluorescence

Frozen sections of the aortic root from mice were incubated at 60 °C for 2 minutes and then fixed in 4% paraformaldehyde for 30 minutes. The samples were subsequently washed with PBS and permeabilized/blocked with 0.1% Triton X-100 in 5% BSA. Next, the primary antibodies, including anti-Xcr1 (BD Biosciences, Cat. 148212) and anti-CD11c (BD Biosciences, Cat. 12-0114-83), were applied in a wet box at 4 °C overnight. After washing with PBST (PBS containing 0.1% Tween-20) for five times, the samples were incubated with secondary antibodies for 1 hour, followed by a 10-minute incubation with DAPI at room temperature (protected from light). The fluorescent signals of the sections were detected using a Nikon C2 confocal microscope (Nikon, Japan).

Morphometric and immunohistochemical analyses

After modeling, the heart and aorta tissue were obtained from mice. The *en face* descending aortas were fixed with 4% paraformaldehyde and subsequently stained with ORO. Additionally, frozen sections of the aortic roots were stained with ORO and H&E to assess the severity of atherosclerosis, and frozen sections of livers were stained with ORO to assess the severity of lipid accumulation, in accordance with our previous study [74]. Paraffin-embedded sections of the aortic root were dewaxed in xylene twice for 10 minutes, rehydrated through descending grades of ethanol, and subjected to sodium citrate-induced antigen retrieval for 10 minutes at 95 °C. The sections were then incubated with 3% H₂O₂ for 10 minutes at room temperature. Following this, the sections were blocked and incubated with primary antibodies, either anti-Xcr1 (NBP1-02343) or CD68 (FB113109). After five washes with DPBS, the sections were incubated with secondary antibodies for 30 minutes and stained with DAB at room temperature. Subsequently, the sections were counterstained with hematoxylin, dehydrated in ethanol, and mounted with neutral balsam. Images were captured using the digital pathology system (Pannoramic MIDI, Hungary), viewed using CaseViewer2.4 software and analyzed using ImageJ software.

Flow cytometry analysis

To obtain the single-cell suspension, aortas, lymph nodes and spleens were collected from each mouse. Spleens were directly dissociated using the Gentle MACS Dissociators (Miltenyi Biotec). Aortas and lymph nodes were cut into small pieces and then dissociated using the Gentle MACS Dissociators. Dissociated tissues digested with digestive enzymes (Collagenase IV, Cat: V900893 from Sigma; DNase I, Cat: DN25 from Sigma; Liberase DH, Cat: 5401054001 from Roche) for 30 minutes at 37 °C using a shaker set to 300 rpm or a heating block. The resulting cell suspension was filtered through 75 µm strainers and subsequently blocked with 2.4G2 for 20 minutes at 4 °C, followed by washing with stain buffer (BD, Cat: 554656). Then samples were stained with monoclonal antibody mixes for 30 minutes at 4 °C followed by washing with stain buffer. Finally, cells were resuspended in stain buffer containing 200 nM Sytox blue (ThermoFisher Scientific, Cat: S34857) and acquired using a flow cytometer (ThermoFisher Scientific, Invitrogen Attune NxT Flow Cytometer), and the FACS data were analyzed using FlowJo software. The following antibodies purchased from BioLegend, BD or eBioscience were used to detect surface markers by flow cytometry: anti-CD45.2 (104) conjugated to eFluor 450, FITC or PE-Cy7; anti-CD45.1 (A20) conjugated to PE or eFluor450; anti-CD45 (30-F11) conjugated to PE; anti-CD11b (M1/70) conjugated to super Bright 600, BUV395 or biotin; anti-CD3e (145-2C11) conjugated to PE-Cy5.5 or PE-Cy7; anti-CD3e (eBio500A2) conjugated to Alexa Fluor700; anti-CD19 (eBio1D3) conjugated to PE-Cy5.5 or biotin; anti-Xcr1 (ZET) conjugated to APC or BV650; anti-Ly6G (1A8) conjugated to Alexa Fluor 700; anti-MHCII (M5/114.15.2) conjugated to APC-eFluor780; anti-CD11c (N418) conjugated to PE or PE-Cy5.5; anti-Siglec H (eBio440c) conjugated to PE-Cy7 or PE; anti-CD8b (eBioH35-17.2) conjugated to eFluor 450; anti-CD44 (IM7) conjugated to BV605; anti-CD4 (RM4-5) conjugated to PE-Cy5.5; anti-CD69 (H1.2F3) conjugated to PE-Cy7; anti-CD62L (MEL-14) conjugated to APC; anti-CD5 (53-7.3) conjugated to APC-eFluor780 or biotin; anti-Siglec F (E50-2440) conjugated to BV421; anti-F4/80 (BM8) conjugated to PE-Cy7; anti-CD64 (X54-5/7.1) conjugated to APC; anti-NK1.1 (PK136) conjugated to Alexa Fluor 700; anti-CD172α (P84) conjugated to FITC.

Single cell RNA sequencing

Eight-week-old male ApoE^{-/-} mice were fed with 20-week ND or HFD. The single cell suspension of spleen, lymph node and aorta were obtained according to the flow cytometry analysis section. Single cell suspension from spleens and lymph nodes were firstly removed the CD5⁺, CD19⁺ and CD11b⁺ cells via SAV conjugated beads. Then two million live cells from spleen and lymph node were stained with mix1 (CD45.2-PE-Cy7, CD11c-PE-Cy5.5, CD172α-FITC, MHCII-APC-eFluor780 and Xcr1-APC), and all cells from aorta were stained with mix2 (CD45.2-PE-Cy7, CD11b-super bright 600, CD11c-PE-Cy5.5, MHCII-FITC, CD3-Alexa Fluor 700, CD19-Alexa Fluor 700, Ly6G-Alexa Fluor 700 and Xcr1-APC) for 30 minutes at 4 °C. After washing with 1 mL FACS buffer two times, cells from different organs were stained with different anti-CD45 antibodies conjugated with barcode oligos from BDTM Ms Single Cell Sample Multiplexing Kit (BD Bioscience, Cat: 633793). Finally, cells were resuspended with Sytox blue and CD11c⁺ MHCII⁺ Xcr1⁺ cells were sorted and subjected to the BD Rhapsody Express system. Then cDNA and sample tag libraries were built with BD Rhapsody whole transcriptome amplification (WTA) reagent kit (BD Bioscience, Cat: 633733, 633773, 633801, 664887) following manufacturer's instructions, and sequenced on an illumine Novaseq 600 sequencer. Pair-end Fastq files of sample tag and WTA data were processed via BD RhapsodyTM analysis pipeline v2.0, and the resultant dataset was mainly analysis using SeqGeqTM software, which contain Lex-BDSMK and Seurat v4.04 plugin components. Raw data and processed data were uploaded into GEO (Accession number: GSE279370).

Statistical analysis

All data were presented as means ± SEM (Standard Error of the Mean), reflecting the average values of biological measurements across multiple samples. Data analysis was performed using GraphPad Prism software (version 8.0). Statistical significance was assessed by both parametric

(unpaired Student's *t*-test) and nonparametric (Mann–Whitney test) methods when comparing two groups. The assumptions of the tests were checked to ensure appropriate application. Significant differences between different groups were set at * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ and **** $P < 0.0001$. Graphical representations of the data will include bar graphs with error bars denoting SEM.

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Additional information

Author Contributions

Lichen Zhang and **Yinming Liang** conceived the project. **Tianhan Li** and **Liaoxun Lu** wrote the manuscript and carried out the majority of the experiments. **Bernard Malissen** and **Lichen Zhang** designed the Xcr1Cre-Gfp knock-in mice and Xcl1 knockout mice. **Juanjuan Qiu**, **Xin Dong**, **Le Yang**, **Kexin He**, and **Yanrong Gu** assisted in the experimental procedures. **Binhui Zhou** performed cell sorting for scRNA sequencing. **Tingting Jia** and **Rong Huang** contributed to data curation and validation. **Toby Lawrence**, **Marie Malissen**, and **Hui Wang** participated in the manuscript review and editing. **Guixue Wang** assisted in the preparation of human samples. **Yinming Liang**, **Lichen Zhang**, and **Tianhan Li** contributed to obtaining the funding for the project.

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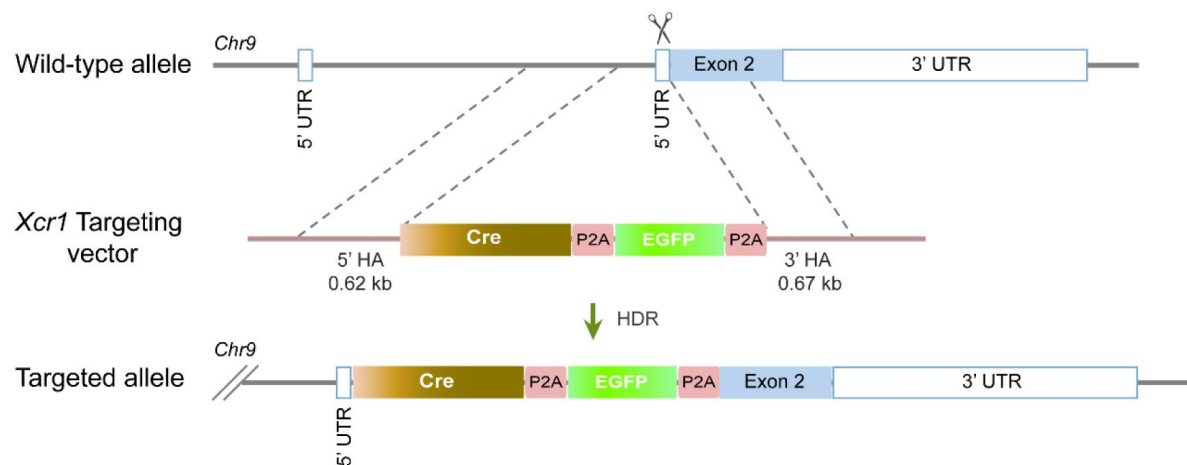


Figure S1.

Schematic diagram illustrating the knock-in of the 5'HA-iCre-P2A-EGFP-P2A-3'HA vector into the mouse *Xcr1* locus.

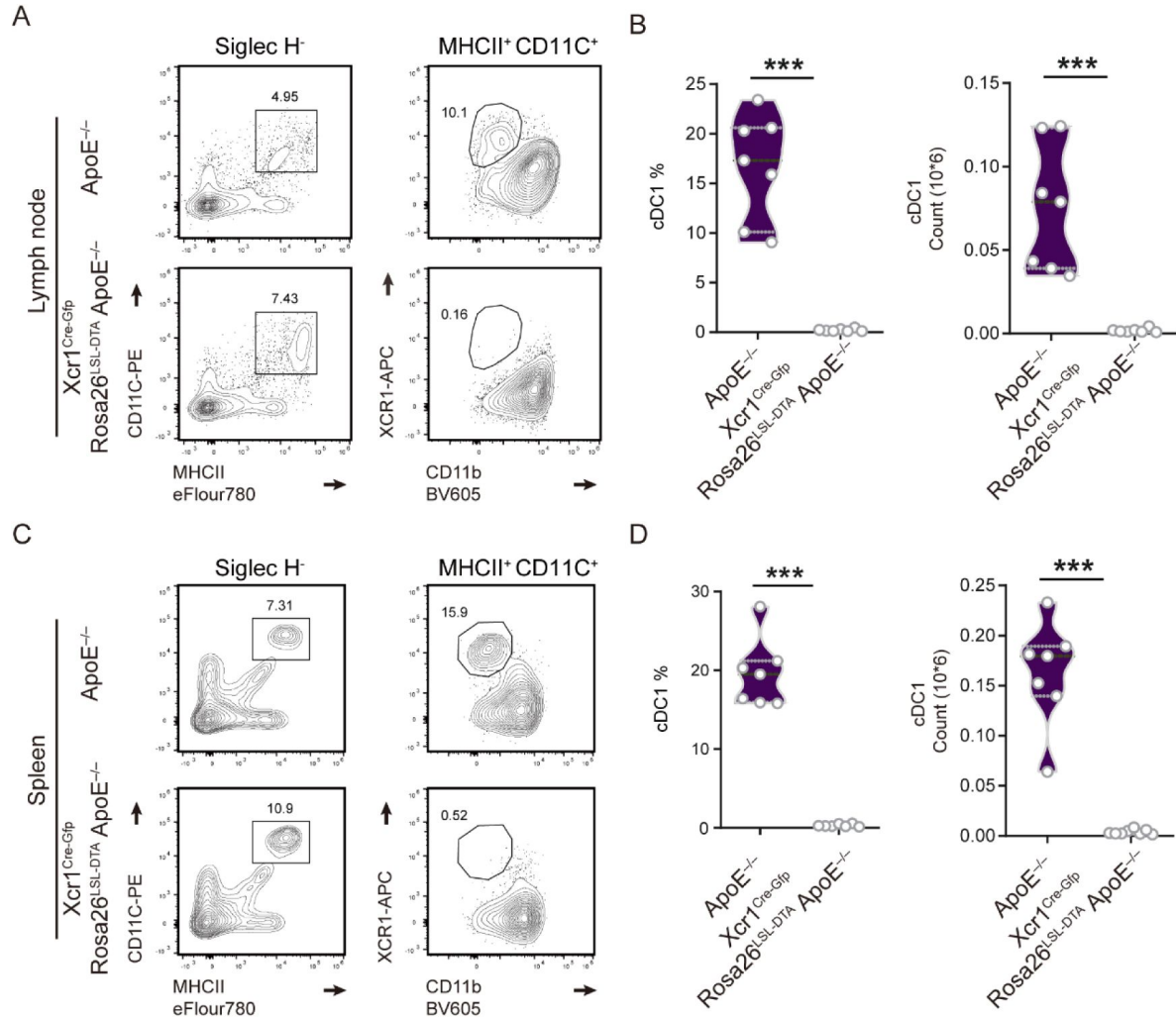


Figure S2.

Flow cytometric analysis of cDC1 cells in the lymph nodes and spleens of ApoE^{-/-} mice and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed a 7-weeks ND.

A, Representative flow cytometric analysis of cDC1 cells in lymph nodes. **B**, Quantification of the frequencies and number of cDC1 cell in lymph nodes. **C**, Representative flow cytometric analysis cDC1 cells in the spleens. **D**, Quantification of the frequencies and number of cDC1 cell in the spleens. ApoE^{-/-} mice and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed with 7-week ND, n = 7. Data represent the mean ± SEM. *** P < 0.001.

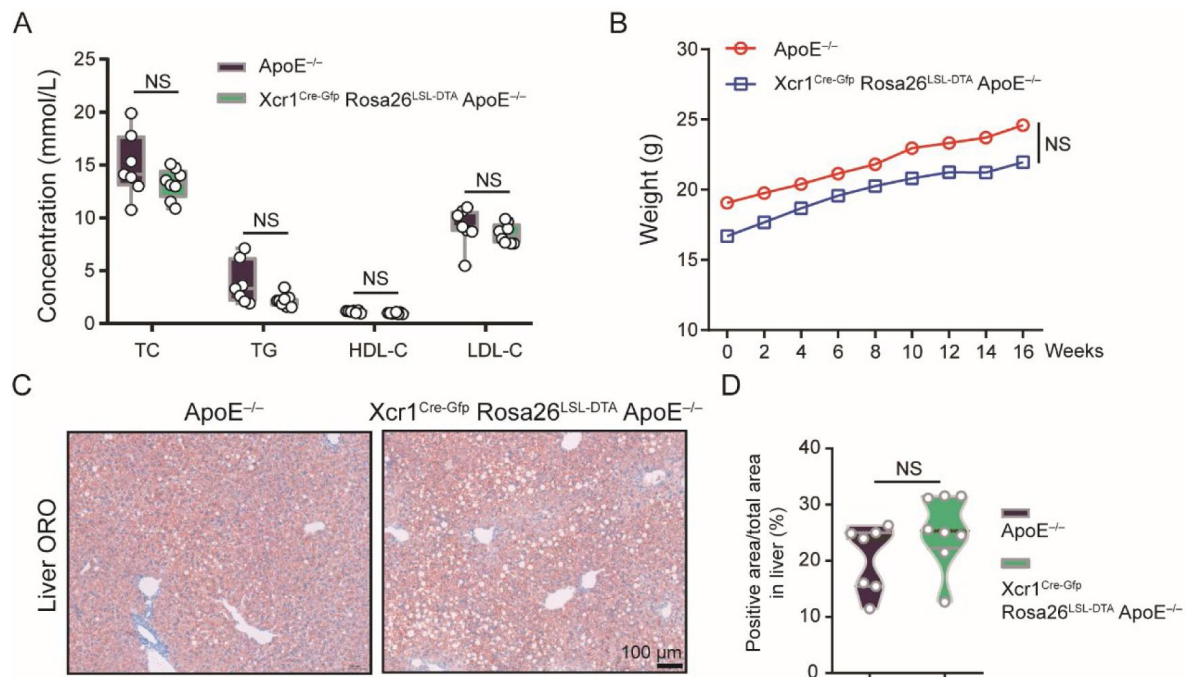


Figure S3.

Lipid profile, body weight and ORO staining of the liver in $ApoE^{-/-}$ mice and $Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}$ mice fed a 16-week HFD.

A, Concentrations of TC, TG, LDL and HDL in the serum. **B**, Body weight measurements of both groups of mice. **C**, ORO staining of liver from each group. Scale bar, 100 μm . **D**, Quantification of ORO positive area in liver. $ApoE^{-/-}$ mice (n =7) and $Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-}$ mice (n =8) were fed on a 16-week HFD to develop atherosclerosis. Data represent the mean $\pm$ SEM. NS, non-significant.

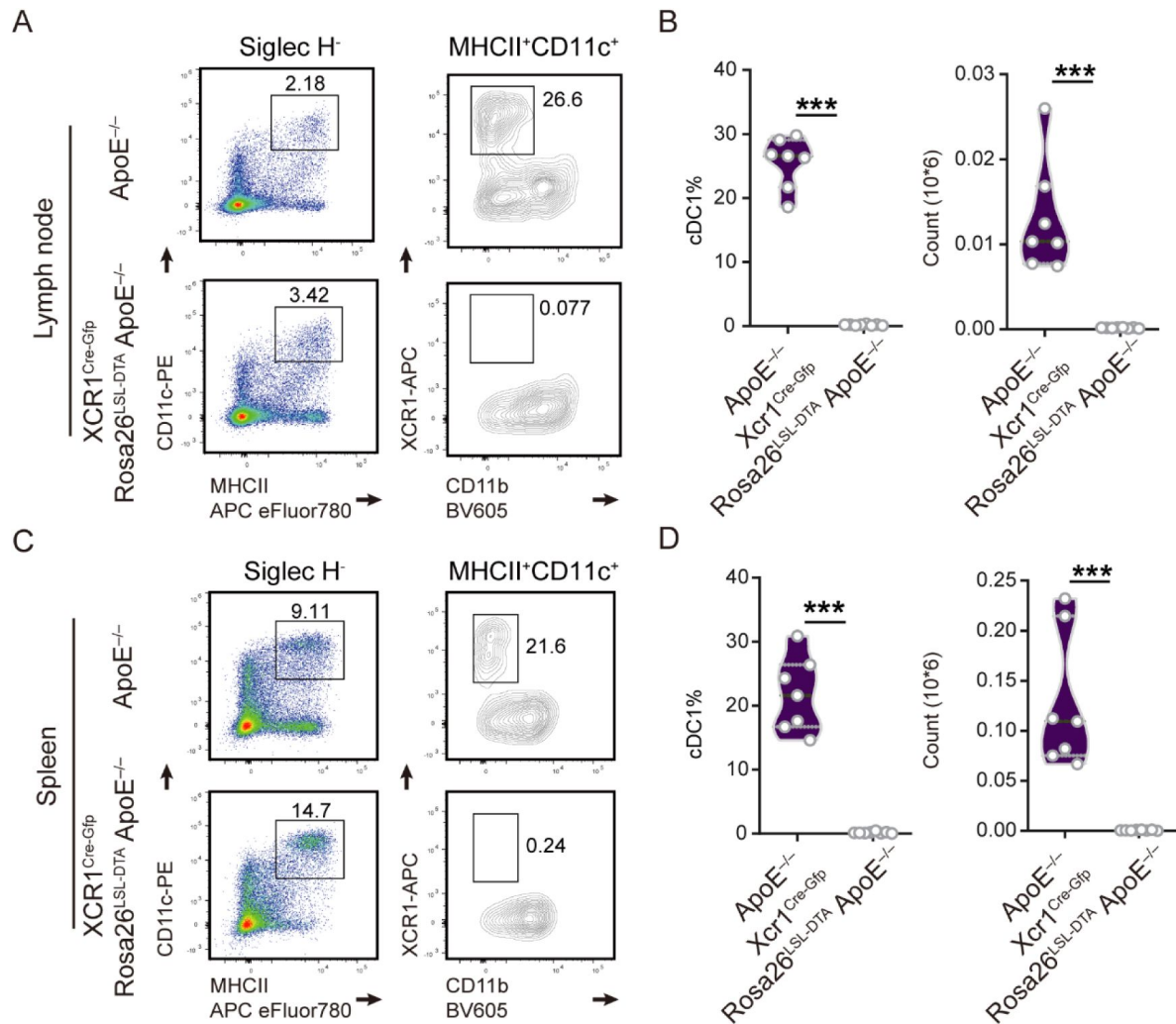


Figure S4.

Flow cytometric analysis of cDC1 cells in the lymph nodes and spleens of ApoE^{-/-} and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed a 16-week HFD.

A, Representative flow cytometric analysis of cDC1 cells in lymph nodes. **B**, Quantification of the frequencies and numbers of cDC1 cells in lymph nodes. **C**, Representative flow cytometric analysis of cDC1 cells in the spleens. **D**, Quantification of the frequencies and numbers of cDC1 cells in the spleens. ApoE^{-/-} mice (n = 7) and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice (n = 8) were fed on a 16-week HFD. Data represent the mean ± SEM. *** P < 0.001.

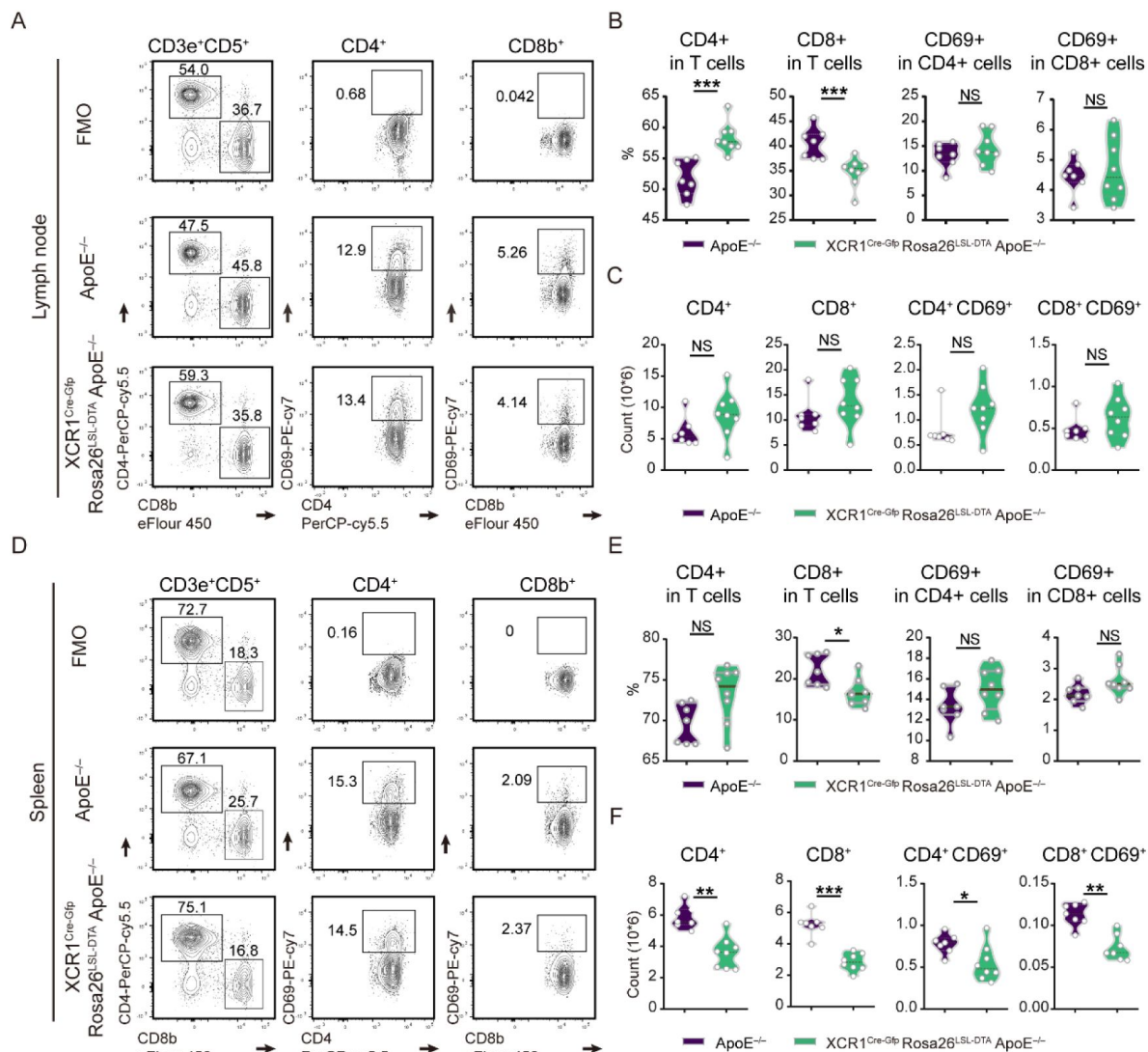


Figure S5.

Flow cytometric analysis of T cells in the lymph nodes and spleens of ApoE^{-/-} mice and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed a 16-week HFD.

A, Representative flow cytometric analysis of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in lymph nodes. **B** and **C**, Quantification of the frequencies and numbers of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in lymph nodes. **D**, Representative flow cytometric analysis and quantification of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in the spleens. **E** and **F**, Quantification of the frequencies and numbers of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in the spleens. ApoE^{-/-} mice (n = 7) and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice (n = 8) were fed on a 16-week HFD. Data represent the mean ± SEM. * P < 0.05, ** P < 0.01, *** P < 0.001, NS, non-significant.

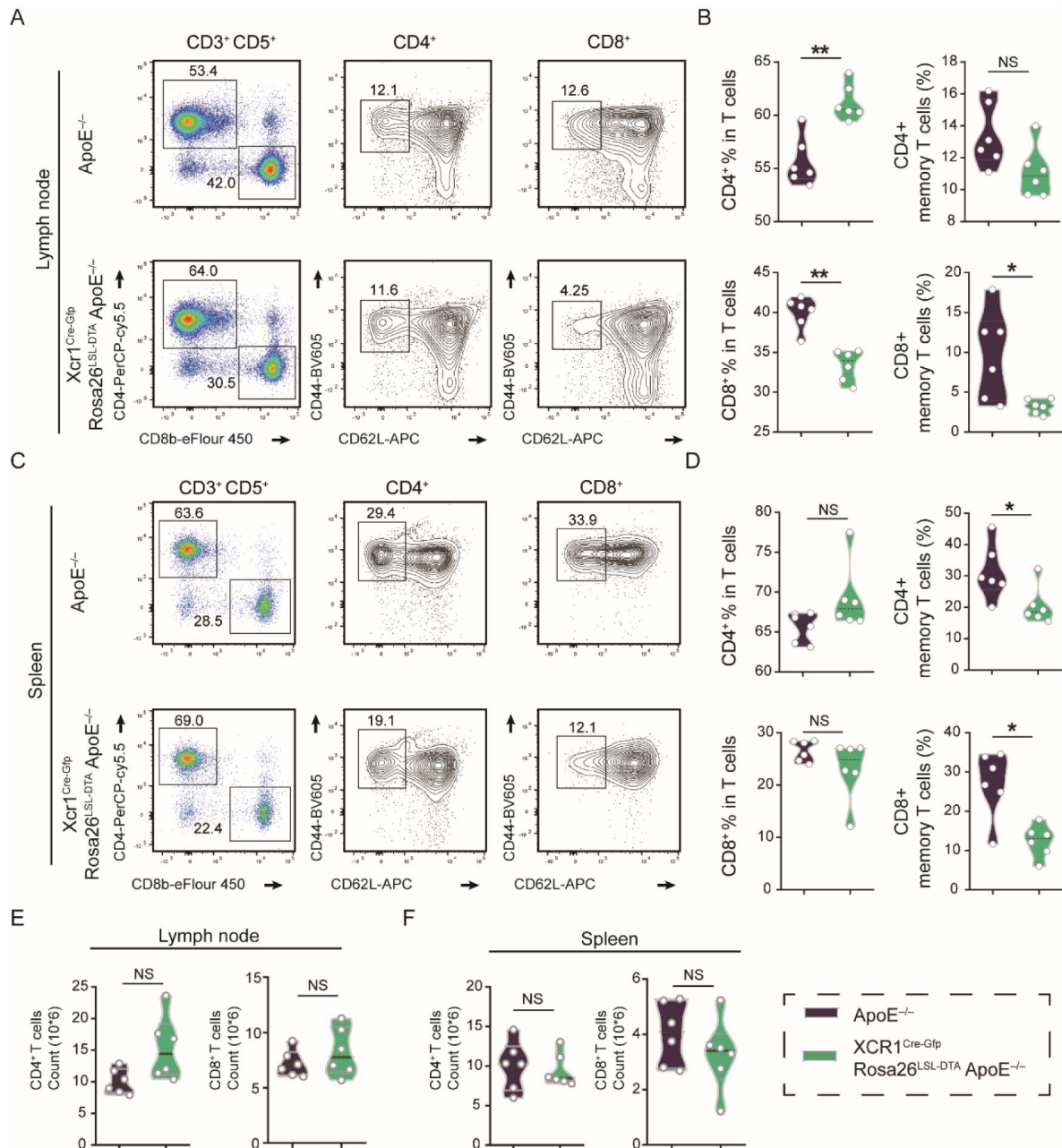


Figure S6.

T cells analysis in the lymph nodes and spleens of ApoE^{-/-} mice and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed a 7-week ND.

A, Representative flow cytometric analysis and quantification of CD4⁺, CD4⁺ memory, CD8⁺ and CD8⁺ memory T cells in lymph nodes. **B**, Quantification of the frequencies of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in the lymph nodes. **C**, Representative flow cytometric analysis and quantification of CD4⁺, CD4⁺ memory, CD8⁺ and CD8⁺ memory T cells in the spleens. **D**, Quantification of the frequencies of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in the spleens. **E** and **F**, Quantification of the numbers of CD4⁺ and CD8⁺ T cells in the lymph nodes and spleens. ApoE^{-/-} mice and Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice fed with a 7-week ND (n = 6). Data represent the mean ± SEM. * P<0.05, ** P<0.01, NS, non-significant.

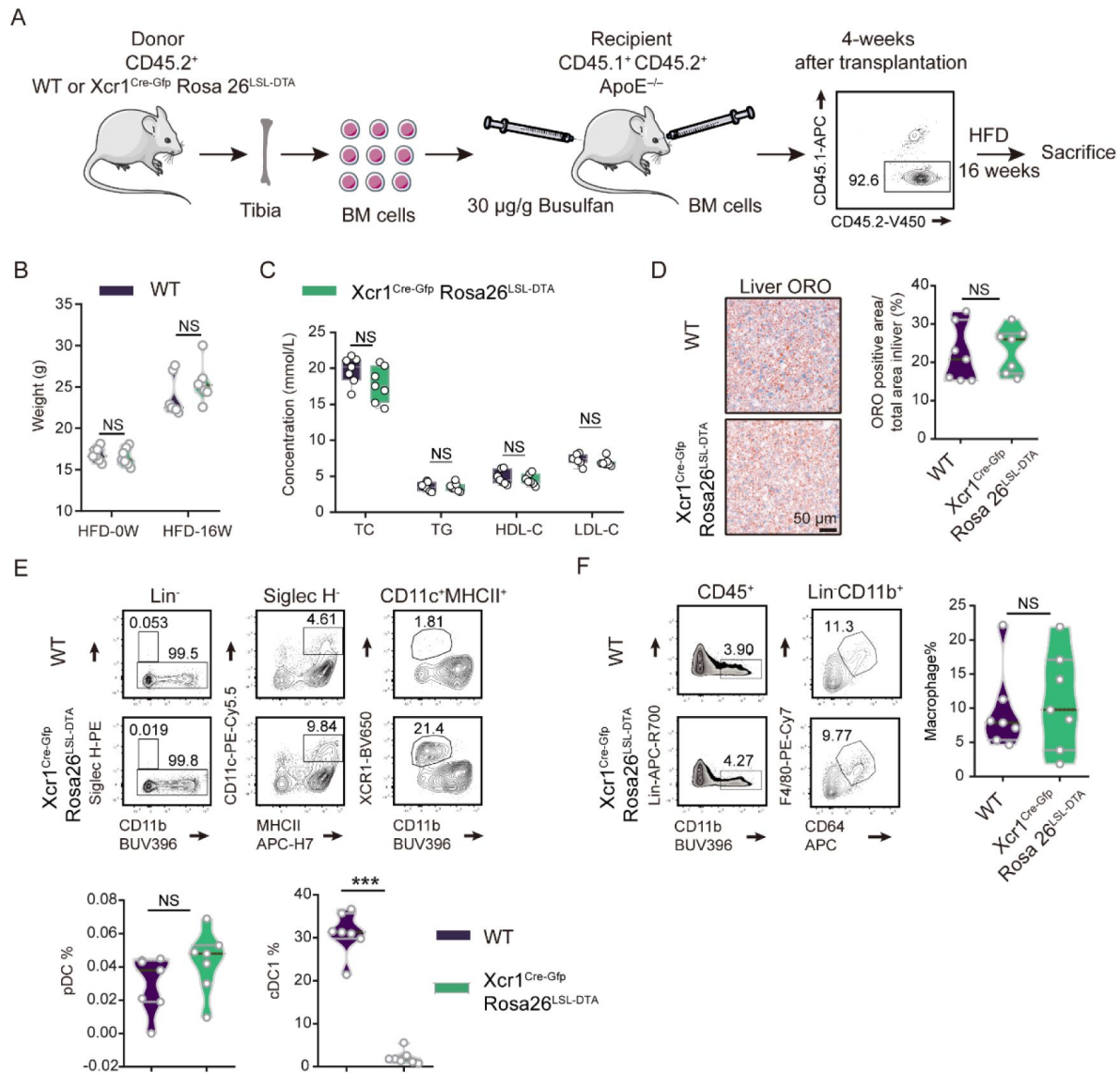


Figure S7.

Depletion of Xcr1⁺ cDC1 cells of bone marrow reduces cDC1 cells without affecting macrophages in atherosclerotic lesion of ApoE^{-/-} mice.

A, Diagram of the bone marrow transfer process. ApoE^{-/-} mice transplanted with bone marrow from WT or Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} donors (n = 7). **B**, Body weight of each group. **C**, Serum concentrations of TC, TG, LDL and HDL. **D**, Representative ORO staining images of liver sections and quantification of the ORO positive area percentage in the liver. Scale bar, 50 µm. **E**, Representative flow cytometric analysis of pDC, cDC1 and cDC2 cells in the aorta, Lin⁻ means CD3⁻CD19⁻Ly6G⁻NK1.1⁻. **F**, Flow cytometric analysis of macrophages in the aorta. Data represent as mean ± SEM. *** P<0.001, NS, non-significant.

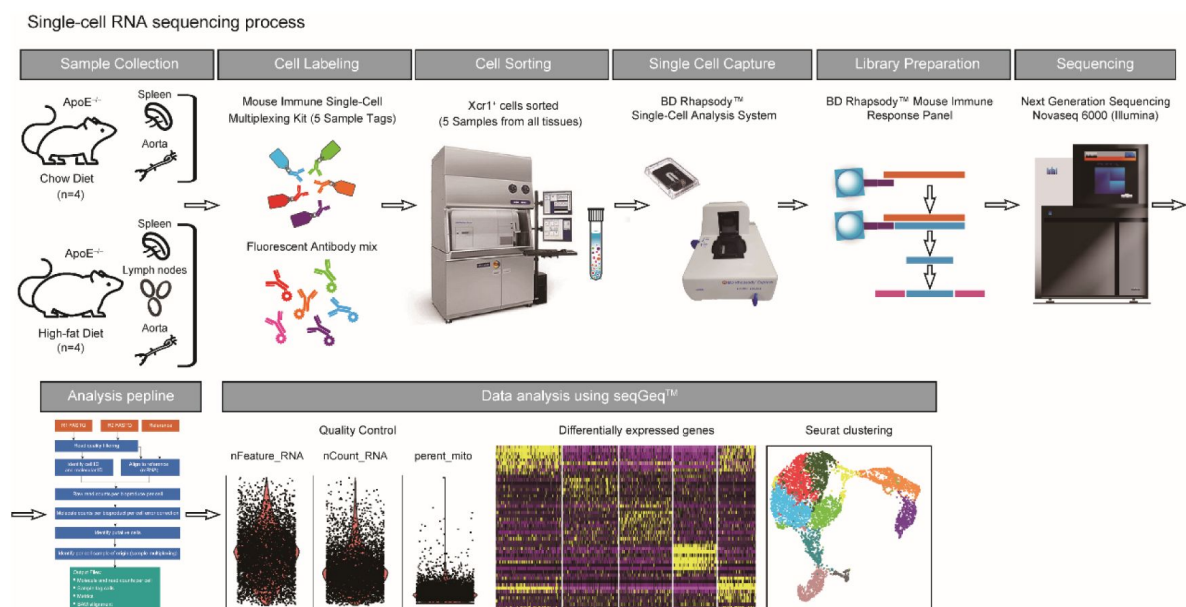


Figure S8.

The single-cell RNA sequencing process.

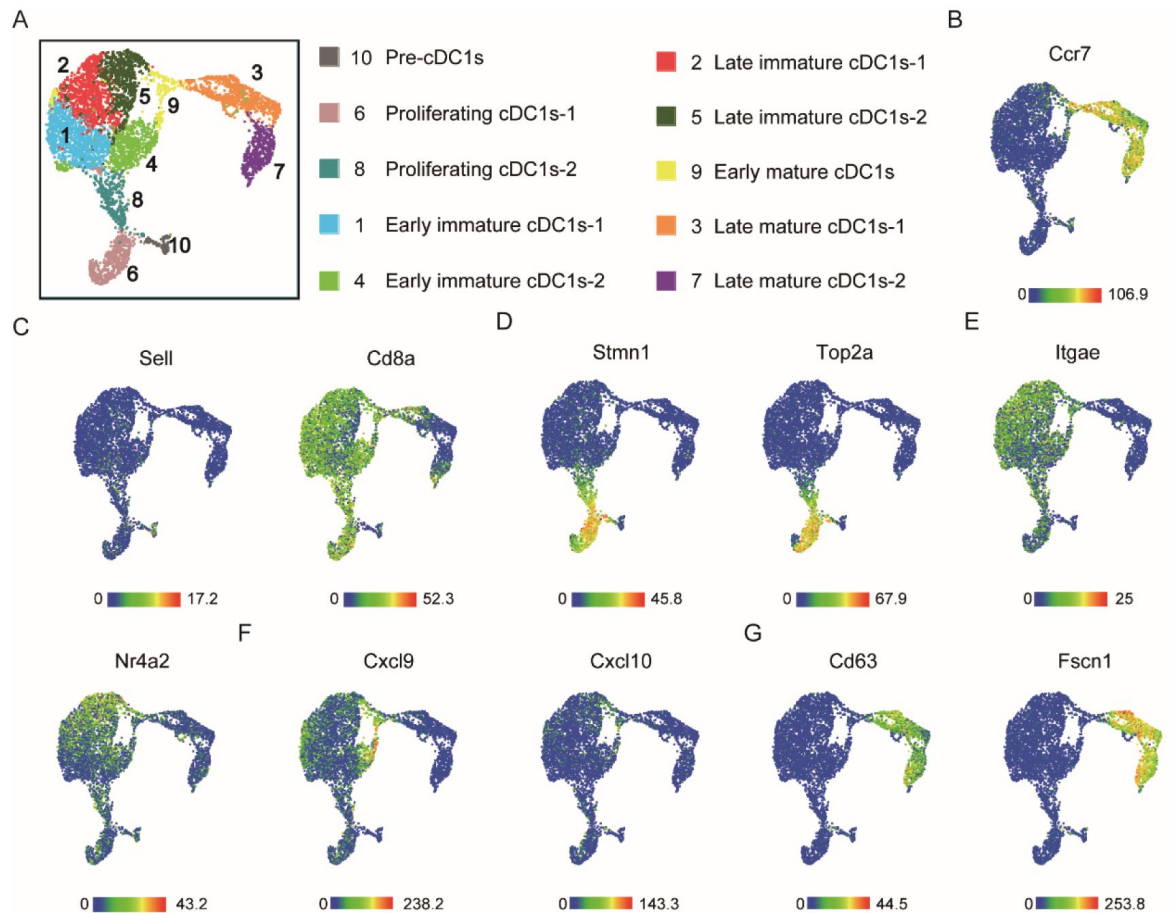


Figure S9.

ScRNA sequencing analysis of the aorta, lymph nodes and spleen in $ApoE^{-/-}$ mice.

A, UMAP plot delineates 10 annotated cell types of $Xcr1^+$ cDC1 cells from aorta, lymph node and spleen in $ApoE^{-/-}$ mice maintained on a HFD or ND for 20 weeks. **B-G**, Heatmap statistic map showing marker genes in different clusters. *Ccr7* is the mature gene marker (B). *Sell*⁺*CD8a*⁻ cDC1s are defined as Pre-cDC1s (C). *Stmn1* and *Top2a* are the proliferating markers (D). *Itgae* and *Nr4a2* genes marked the transition from early to late immature cDC1s (E). *Cxcl9* and *Cxcl10* genes marked the transition from late immature to early mature cDC1s (F). *Cd63* and *Fscn1* are the late mature markers (G).

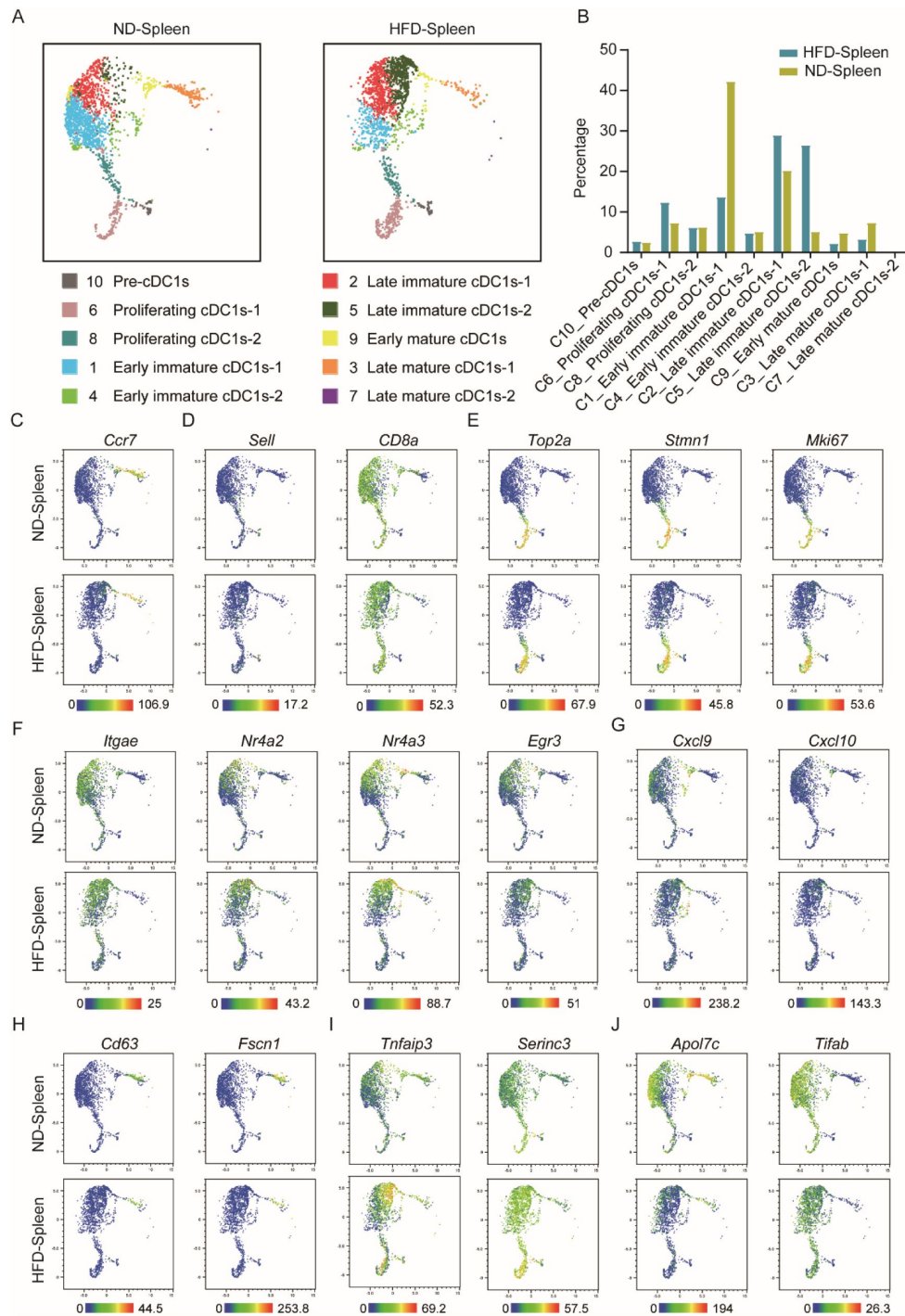


Figure S10.

Single-cell RNA sequencing of splenic cDC1s from both the HFD and ND groups.

A, UMAP plot of splenic cDC1s showing annotated clusters. **B**, The proportions of cDC1 maturation states across diet groups. **C-H**, Heatmap statistic map depicting representative markers defining cDC1 maturation clusters. *Ccr7* is the mature gene marker (C). *Sell*⁺*CD8a*⁺ cDC1s are defined as Pre-cDC1s (D). *Top2a*, *Stmn1* and *Mki67* are the proliferating markers (E). *Itgae*, *Nr4a2*, *Nr4a3* and *Egr3* genes marked the transition from early to late immature cDC1s (F). *Cxcl9* and *Cxcl10* genes marked the transition from late immature to early mature cDC1s (G). *Cd63* and *Fscn1* are the late mature markers (H). **I-J**, Heatmap statistic map of upregulated genes (*Tnfrsf3* and *Serinc3*), and downregulated genes (*Apol7c* and *Tifab*) in the HFD spleen compared to the ND spleen.

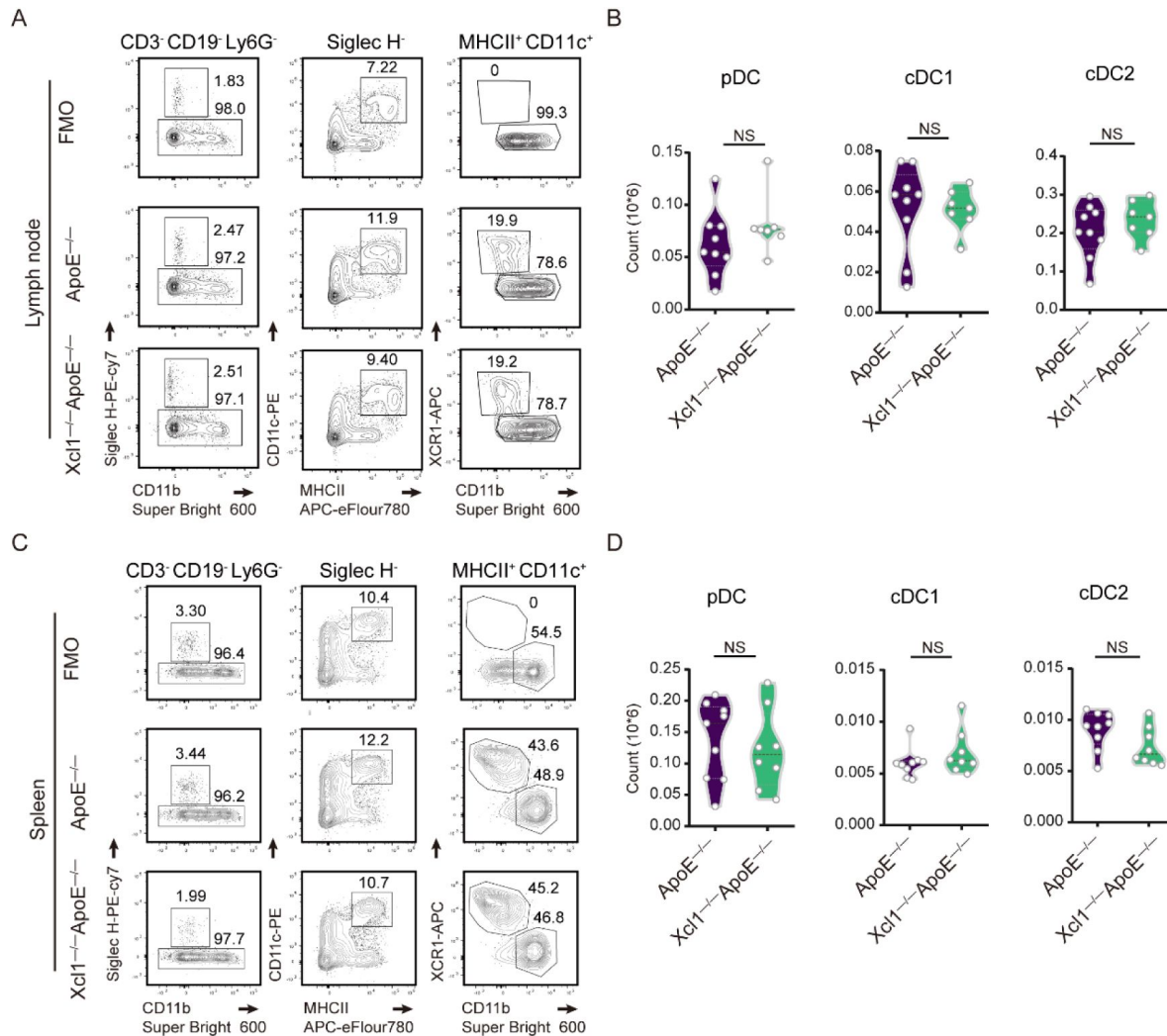


Figure S11.

DC cells analysis in lymph nodes and spleens of ApoE^{-/-} and Xcl1^{-/-} ApoE^{-/-} mice fed with 19-week HFD.

A, Representative flow cytometric analysis and quantification of pDC, cDCs, cDC1 and cDC2 cells in lymph nodes. **B**, Quantification of absolute counts of pDC, cDCs, cDC1 and cDC2 cells in lymph nodes. **C**, Representative flow cytometric analysis of pDC, cDC1 and cDC2 cells in spleens. **D**, Quantification of absolute counts of pDC, cDCs, cDC1 and cDC2 cells in spleens. ApoE^{-/-} (n = 9) and Xcl1^{-/-} ApoE^{-/-} mice (n = 7 or 8) fed a HFD for 19 weeks. Data represent the mean ± SEM. NS, non-significant.

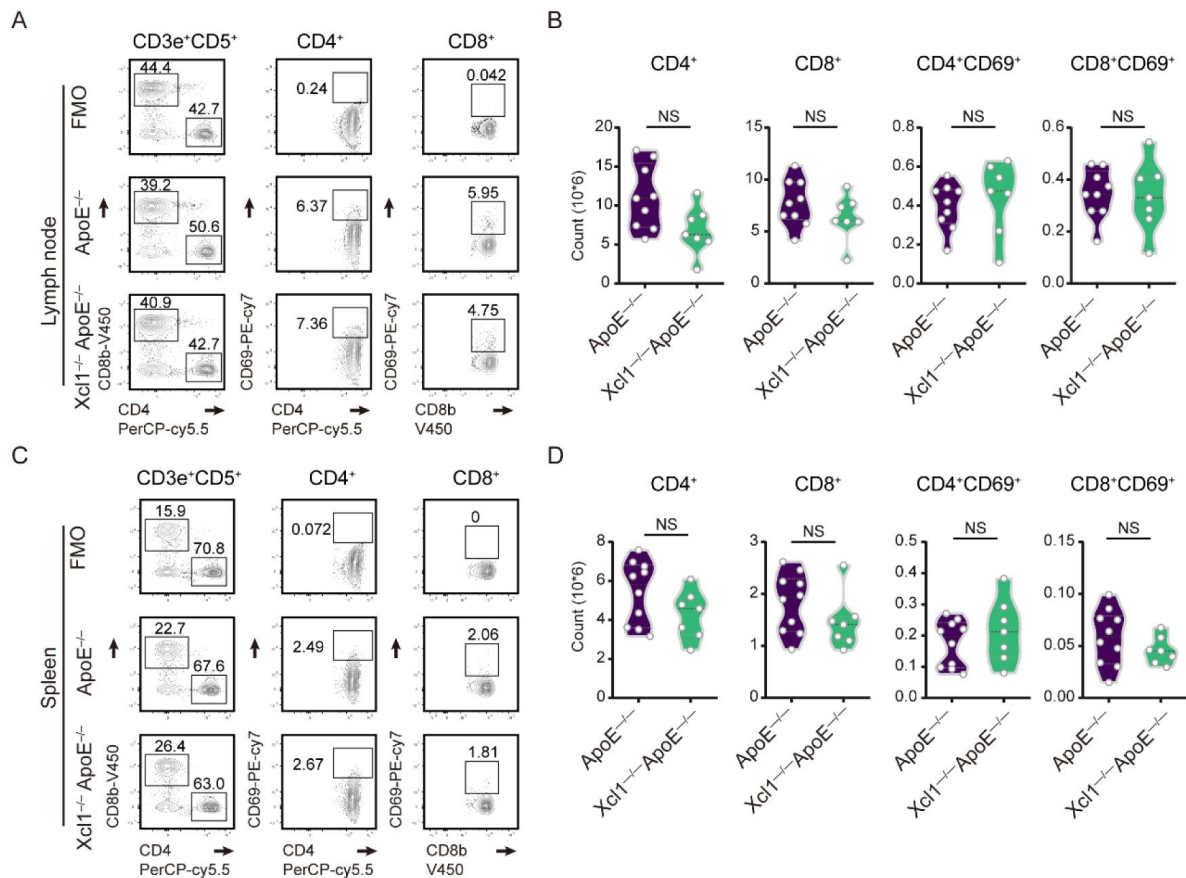


Figure S12.

T cells analysis in lymph node and spleen of ApoE^{-/-} and Xcl1^{-/-} ApoE^{-/-} mice fed with 16-week HFD.

A and **B**, Representative flow cytometric analysis and quantification of absolute counts of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in lymph nodes (ApoE^{-/-}, n = 9; Xcl1^{-/-} ApoE^{-/-}, n = 7). **C** and **D**, Representative flow cytometric analysis and quantification of absolute counts of CD4⁺, CD8⁺, CD4⁺CD69⁺ and CD8⁺CD69⁺ T cells in spleens (ApoE^{-/-}, n = 10; Xcl1^{-/-} ApoE^{-/-}, n = 7). Data represent the mean ± SEM. NS, non-significant.

[GeneSet]	
\$SchemaVersion 1	
\$GeneSetName	Seurat Marker Features RunID: ZEE2
\$GeneSetDescription	Generated in the Seurat analysis pipeline. Fold-Change > 5.2605272115588; q-Val (FDR) < 1

[Genes]	Log2 Fold-Change	q-Value	Cluster
Hspa1b	42.52120302	9.50E-177	1
Dnajb1	20.81074839	2.92E-90	1
GlrX	13.83830441	3.45E-48	1
Hes1	10.52069041	4.85E-44	1
Dnaja1	25.13842471	2.68E-41	1
Hspd1	12.1502349	7.64E-13	1
Lpar6	10.56634729	1.18E-11	1
Cxcr4	13.53264732	0.007875697	1
Cldnd1	33.56256056	1	1
Sat1	16.01666556	1	1
Stk17b	9.341881929	1.00E-120	2
Gpr171	8.036975348	6.90E-78	2
Cd36	37.33127974	3.01E-59	2
Plekho2	9.679954684	1.26E-51	2
Cd14	15.11839556	1.73E-20	2
Smim3	9.359129334	1.29E-10	2
Uck2	8.412740797	1.12E-09	2
Rraga	8.278727722	1.62E-06	2
Nsd1	8.142796627	8.74E-06	2
Sgk1	9.19132226	0.628752449	2
Ccl5	Inf	0	3
Epsti1	106.9899766	0	3
AW112010	37.55982075	0	3
Tspan3	29.08304265	0	3
Serpinb6b	43.04386652	2.81E-211	3
Vim	63.2513173	2.29E-95	3
Ccl22	178.6669207	6.38E-81	3
Tagln2	43.81517158	9.66E-79	3
Resf1	28.47128144	4.78E-63	3
Prdx1	34.18432187	4.61E-36	3
Slfn5	33.17542571	1.81E-130	4
Ifi44	11.07741437	1.18E-80	4
Irf7	18.03968574	4.53E-78	4
Stat1	11.19127012	2.20E-70	4

Supplementary table 1

The top ten high-expressed marker genes in ten clusters

Ly6a	71.77936127	4.96E-35	4
Isg15	13.81873278	1.38E-27	4
Myd88	13.02295703	1.08E-11	4
Ifitm3	10.60216615	0.000403532	4
Nr4a2	12.68639324	1	4
Eprs	32.65659433	1	4
Ifrd1	13.16353946	7.56E-128	5
Nlrp3	14.45121644	2.01E-99	5
Tnfsf9	17.66324735	1.68E-76	5
Gpr65	23.75629403	3.36E-57	5
Itga4	14.48761702	1.91E-53	5
Aff1	15.6278723	6.96E-45	5
Lyz2	16.00040211	5.23E-30	5
Chd7	13.03464538	3.32E-27	5
Ccr12	15.03814065	6.89E-19	5
Anxa1	17.1352024	0.000648149	5
Ube2c	34.35930797	0	6
Tuba1b	33.6015791	0	6
Hmgb2	21.96553285	0	6
Arl6ip1	20.79152855	0	6
Mki67	20.52766974	0	6
Tubb5	14.2678514	0	6
Cenpe	10.98235375	0	6
Top2a	19.59760586	2.19E-271	6
Cpox	12.47616675	3.47E-21	6
Hsph1	23.51902433	1	6
Fabp5	22.30871744	8.14E-283	7
2610528A11Rik	66.90179888	1.03E-218	7
S100a4	21.28583493	1.45E-149	7
Tmem123	69.90785515	4.20E-133	7
Mt1	52.98793574	4.91E-90	7
Gpr183	20.03728313	3.03E-84	7
S100a6	25.56962085	6.88E-76	7
Ctsc	23.64358551	2.62E-46	7
Cdk2ap2	39.81012543	9.71E-37	7
Clu	25.18396035	1.86E-20	7
Pena	8.142821346	5.22E-182	8
Mcm7	5.358022282	3.65E-177	8
Mcm5	6.528065009	7.55E-166	8
Lig1	6.681288261	1.09E-157	8
Dut	5.260527212	6.64E-133	8
Paics	6.750359239	1.54E-111	8
Set	9.064414218	3.85E-87	8

Supplementary table 1 (continued)

Stmn1	6.753546782	8.71E-18	8
Tmem14c	7.079252859	1.11E-11	8
Npm3	7.295805042	1.78E-10	8
Nampt	14.01602029	5.52E-94	9
Serpina3g	21.08183631	2.44E-40	9
Sqstm1	35.71080346	2.69E-35	9
Nr4a3	46.8592915	2.27E-30	9
Prkcd	14.53860632	1.53E-28	9
Bcl2l11	16.47721576	2.90E-18	9
Cd274	20.28939966	4.91E-12	9
Ifi207	17.33536354	1.81E-09	9
Cdkn1a	14.73834176	6.97E-08	9
Gm19951	16.14302152	1.04E-06	9
Ets1	24.68664964	1.25E-263	10
Il2rb	40.60134549	3.11E-168	10
Nkg7	26.174639	5.46E-153	10
Xcl1	179.0903763	4.57E-147	10
Ighm	356.1167175	1.40E-138	10
Cd7	51.65059839	4.00E-68	10
Nrgn	63.22973626	6.64E-60	10
Igkc	Inf	1.11E-59	10
Ccnd2	22.7688043	1.11E-11	10
Odc1	79.11780903	1.83E-09	10

Supplementary table 1 (continued)

[GeneSet] \$SchemaVersion 1 \$GeneSetName up regulated genes in HFD-Aorta vs HFD-Spleen \$GeneSetDescription				
[Genes]	fold_change(HFD-Aorta vs HFD-Spleen)	q-value(HFD-Aorta vs HFD-Spleen)	HFD-Aorta	HFD-Spleen
Ccr2	2.868	2.08E-13	2.7647	0.3127
Cldnd1	2.09622	4.98E-05	3.4706	1.1327
Ifi203	2.03954	0.000476302	1.5735	0.2618
Mndal	2.38451	4.03E-08	2.3677	0.4123
3-Sep	2.07201	2.26E-07	4.9706	1.8816
Wfdc17	2.01749	0.0194669	1.1765	0.0788

[GeneSet] \$SchemaVersion 1 \$GeneSetName up regulated genes in HFD-Aorta vs HFD-Lymph nodes \$GeneSetDescription				
[Genes]	fold_change(HFD-Aorta vs HFD-Lymph nodes)	q-value(HFD-Aorta vs HFD-Lymph nodes)	HFD-Aorta	HFD-Lymph nodes
Ccr2	2.47261	2.08E-13	2.7647	0.5226
Cd24a	2.04889	1.39E-10	8.2206	3.5003
Ckb	2.43031	2.08E-13	4.2059	1.1421
Cldnd1	2.85069	2.08E-13	3.4706	0.5682
Cyth4	2.11238	2.08E-13	3.0588	0.9214
Dnaja1	3.22678	1.02E-10	6.7353	1.3972
Dnajb1	5.16719	2.48E-09	8.0735	0.756
Fgfr1	2.25182	2.08E-13	1.5441	0.1298
Hsp90aa1	4.50681	3.09E-08	21.441	3.9794
Hsp90ab1	2.13951	1.05E-05	29.074	13.056
Hspa1a	7.12033	2.08E-13	36.779	4.3059
Hspa1b	9.11898	2.08E-13	38.677	3.351
Hspa5	2.68952	2.39E-08	23.059	7.9454
Hspd1	2.33134	0.0275939	3.1471	0.7788
Hspe1	2.16981	1.12E-05	6.1029	2.2735
Hsph1	5.51596	1.10E-06	6.75	0.405
Ifitm2	2.29707	1.85E-11	3.3971	0.9142
Ifitm3	2.23302	0.000253682	2.0441	0.3632
Lyz2	2.1711	2.08E-13	4.0147	1.3098
Naaa	2.25383	2.08E-13	19.177	7.9521

Supplementary table 2

Up regulated genes in Xcr1⁺ cDC1 cells from aorta vs spleen, and aorta vs lymph node in ApoE-deficient mice fed a HFD for 20 weeks

Supplementary table 2 (continued)

Nr4a2	2.01215	0.0210806	2.5882	0.7833
Olfm1	2.03054	2.08E-13	4.2353	1.5783
Picalm	2.09067	8.77E-08	3.8382	1.3142
Plac8	2.56781	2.08E-13	4.8824	1.2908
Rac2	2.2199	2.08E-13	1.9706	0.3382
Rsrp1	2.13894	2.08E-13	3.3529	1.0351
3-Sep	2.06936	2.08E-13	4.9706	1.8852
Tiparp	2.05071	0.00113877	2.2206	0.5705
Wfdc17	2.08139	6.31E-06	1.1765	0.0457
Zfand2a	2.08069	0.040016	1.2059	0.0602

Supplementary table 3

The sequences of the sgRNA used for generation of *Xcr1*^{Cre-Gfp} knock-in mice and *Xcl1* knockout mice

sgRNA		
sgRNA name	sequence (5' > 3')	Purpose
sgXcr1-KI	AGGCTGTAGAGGACTCCATC <i>TGG</i>	Generation of <i>Xcr1</i> ^{Cre-Gfp} knock-in mice
sgXcl1-1	TTGCGAGGGGCTTTCGGGGT <i>GGG</i>	Generation of <i>Xcl1</i> knockout mice
sgXcl1-2	CAAGCTGAACTTAAGAACGA <i>TGG</i>	

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Reviewer #1 (Public review):

Summary:

In this study by Li et al., the authors re-investigated the role of cDC1 for atherosclerosis progression using the ApoE model. First, the authors confirmed the accumulation of cDC1 in atherosclerotic lesions in mice and humans. Then in order to examine the functional relevance of this cell type, the authors developed a new mouse model to selectively target cDC1. Specifically, they inserted the Cre recombinase directly after the start codon of endogenous XCR1 gene, thereby avoiding off-target activity. Following validation of this model, the authors crossed it with ApoE-deficient mice and found a striking reduction of

aortic lesions (numbers and size) following high fat diet. The authors further characterized the impact of cDC1 depletion on lesional T cells and their activation state. Also, they provide in-depth transcriptomic analyses of lesional in comparison to splenic and nodal cDC1. These results imply cellular interactions between lesion T cells and cDC1. Finally, the authors show that the chemokine XCL1, which is produced by activated CD8 T cells (and NK cells) plays a key role for the interaction with XCR1-expressing cDC1 and particularly for the atherosclerotic disease progression.

Strengths:

The surprising results on XCL1 represent a very important gain in knowledge. The role of cDC1 is clarified with a new genetic mouse model.

Comments on revised version:

The authors have addressed my concerns in the revised version of this manuscript.

<https://doi.org/10.7554/eLife.107742.2.sa2>

Reviewer #2 (Public review):

This study investigates the role of cDC1 in atherosclerosis progression using Xcr1Cre-Gfp Rosa26LSL-DTA ApoE^{-/-} mice. The authors demonstrate that selective depletion of cDC1 reduces atherosclerotic lesions in hyperlipidemic mice. While cDC1 depletion did not alter macrophage populations, it suppressed T cell activation (both CD4⁺ and CD8⁺ subsets) within aortic plaques. Further, targeting the chemokine Xcl1 (ligand of Xcr1) effectively inhibits atherosclerosis. The manuscript is well-written, and data are clearly presented. The data provided in the article can well support the author's conclusion.

Comments on revised version:

The authors have addressed all previous concerns and made appropriate revisions to the data. I have no further questions.

<https://doi.org/10.7554/eLife.107742.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

In this study by Li et al., the authors re-investigated the role of cDC1 for atherosclerosis progression using the ApoE model. First, the authors confirmed the accumulation of cDC1 in atherosclerotic lesions in mice and humans. Then, in order to examine the functional relevance of this cell type, the authors developed a new mouse model to selectively target cDC1. Specifically, they inserted the Cre recombinase directly after the start codon of the endogenous XCR1 gene, thereby avoiding off-target activity. Following validation of this model, the authors crossed it with ApoE-deficient mice and found a striking reduction of aortic lesions (numbers and size) following a high-fat diet. The authors further characterized the impact of cDC1 depletion on lesional T cells and their activation state. Also, they provide in-depth transcriptomic analyses of lesional in comparison to splenic and nodal cDC1. These results imply cellular interactions between

lesion T cells and cDC1. Finally, the authors show that the chemokine XCL1, which is produced by activated CD8 T cells (and NK cells), plays a key role in the interaction with XCR1-expressing cDC1 and particularly in the atherosclerotic disease progression.
Strengths:

The surprising results on XCL1 represent a very important gain in knowledge. The role of cDC1 is clarified with a new genetic mouse model.

Thank you

Weaknesses:

My criticism is limited to the analysis of the scRNAseq data of the cDC1. I think it would be important to match these data with published data sets on cDC1. In particular, the data set by Sophie Janssen's group on splenic cDC1 might be helpful here (PMID: 37172103; https://www.single-cell.be/spleen_cDC_homeostatic_maturation/datasets/cdc1). It would be good to assign a cluster based on the categories used there (early/late, immature/mature, at least for splenic DC).

Thank you very much for your help. Using the scRNA seq data of Xcr1⁺ cDC1 sorted from ApoE^{-/-} mice, we re-annotated the populations, following the methodology proposed by Sophie Janssen's group. These results are presented in Figure S9 and Figure S10 and described in detail in the Results and Discussion section.

Please refer to the Results section from line 264 to 284: “Using the scRNA seq data of Xcr1⁺ cDC1 sorted from hyperlipidemic mice, we annotated the 10 populations as shown in Figure S9A, following the methodology from a previous study [41]. Ccr7⁺ mature cDC1s (Cluster 3, 7 and 9) and Ccr7⁻ immature cDC1s (remaining clusters) were identified across cDC1 cells sorted from aorta, spleen and lymph nodes (Figure S9B). Further stratification based on marker genes reveals that Cluster 10 is the pre-cDC1, with high expression level of CD62L (Sell) and low expression level of CD8a (Figure S9C). Cluster 6 and 8 are the proliferating cDC1s, which express high level of cell cycling genes Stmn1 and Top2a (Figure S9D). Cluster 1 and 4 are early immature cDC1s, and cluster 2 and 5 are late immature cDC1s, according to the expression pattern of Itgae, Nr4a2 (Figure S9E). Cluster 9 cells are early mature cDC1s, with elevated expression of Cxcl9 and Cxcl10 (Figure S9F). Cluster 3 and 7 as late mature cDC1s, characterized by the expression of Cd63 and Fcscn1 (Figure S9G). As shown in Figure 5C and Figure S9, the 10 populations displayed a major difference of aortic cDC1 cells that lack in pre-cDC1s (cluster 10) and mature cells (cluster 3, 7 and 9). Interestingly, in hyperlipidemic mice splenic cDC1 possess only Cluster 3 as the late mature cells while the lymph node cDC1 cells have two late mature populations namely Cluster 3 and Cluster 7. In further analysis, we also compared splenic cDC1 cells from HFD mice to those from ND mice. As shown in Figure S10, HFD appears to impact early immature cDC1-1 cells (Cluster 1) and increases the abundance of late immature cDC1 cells (Cluster 2 and 5), regardless of the fact that all 10 populations are present in two origins of samples. We also found that Tnfaip3 and Serinc3 are among the most upregulated genes, while Apol7c and Tifab are downregulated in splenic cDC1 cells sorted from HFD mice”.

Please refer to the Discussion section from line 380 to 385: “Based on the maturation analysis of the cDC1 scRNA seq data [41], our findings suggest that the aortic cDC1 cells display a major difference from those of spleen and lymph nodes by lacking the mature clusters, whereas lymph node cDC1 cells contain an additional Fabp5⁺ S100a4⁺ late mature Cluster. Our results also suggest that hyperlipidemia contributes to alteration in early immature cDC1 and in the abundance of late immature cDC1 cells, which was associated with dramatic change in gene expression of Tnfaip3, Serinc3, Apol7c and Tifab”.

Reviewer #2 (Public review):

This study investigates the role of cDC1 in atherosclerosis progression using Xcr1Cre-Gfp Rosa26LSL-DTA ApoE^{-/-} mice. The authors demonstrate that selective depletion of cDC1 reduces atherosclerotic lesions in hyperlipidemic mice. While cDC1 depletion did not alter macrophage populations, it suppressed T cell activation (both CD4⁺ and CD8⁺ subsets) within aortic plaques. Further, targeting the chemokine Xcl1 (ligand of Xcr1) effectively inhibits atherosclerosis. The manuscript is well-written, and the data are clearly presented. However, several points require clarification:

(1) In Figure 1C (upper plot), it is not clear what the Xcr1 single-positive region in the aortic root represents, or whether this is caused by unspecific staining. So I wonder whether Xcr1 single-positive staining can reliably represent cDC1. For accurate cDC1 gating in Figure 1E, Xcr1+CD11c⁺ co-staining should be used instead.

The observed false-positive signal in the wavy structures within immunofluorescence Figure 1C (upper panel) results from the strong autofluorescence of elastic fibers, a major vascular wall component (alongside collagen). This intrinsic property of elastic fibers is a well-documented confounder in immunofluorescence studies [A, B].

In contrast, immunohistochemistry (IHC) employs an enzymatic chromogenic reaction (HRP with DAB substrate) that generates a brown precipitate exclusively at antigen-antibody binding sites. Importantly, vascular elastic fibers lack endogenous enzymatic activity capable of catalyzing the DAB reaction, thereby preventing this source of false positivity in IHC.

Given that Xcr1 is exclusively expressed on conventional type 1 dendritic cells [C], and considering that IHC lacks the multiplexing capability inherent to immunofluorescence for antigen co-localization, single-positive Xcr1 staining reliably identifies cDC1s in IHC results.

[A] König, K et al. "Multiphoton autofluorescence imaging of intratissue elastic fibers." *Biomaterials* vol. 26,5 (2005): 495-500. doi:10.1016/j.biomaterials.2004.02.059

[B] Andreasson, Anne-Christine et al. "Confocal scanning laser microscopy measurements of atherosclerotic lesions in mice aorta. A fast evaluation method for volume determinations." *Atherosclerosis* vol. 179,1 (2005): 35-42. doi:10.1016/j.atherosclerosis.2004.10.040

[C] Dorner, Brigitte G et al. "Selective expression of the chemokine receptor XCR1 on cross-presenting dendritic cells determines cooperation with CD8⁺ T cells." *Immunity* vol. 31,5 (2009): 823-33. doi:10.1016/j.immuni.2009.08.027

(2) Figure 4D suggests that cDC1 depletion does not affect CD4⁺/CD8⁺ T cells. However, only the proportion of these subsets within total T cells is shown. To fully interpret effects, the authors should provide:

(a) Absolute numbers of total T cells in aortas.

(b) Absolute counts of CD4⁺ and CD8⁺ T cells.

Thanks for your suggestions. We agree that assessing both proportions and absolute numbers in Figure 4 provides a more complete picture of the effects of cDC1 depletion on T cell populations. Furthermore, we also add the absolute count of cDC1 cells and total T cells, and CD44 MFI (mean fluorescence intensity) in CD4⁺ and CD8⁺ T cells in Figure 4, and supplemented corresponding textual descriptions in the revised manuscript.

Please refer to the Results section from line 183 to 187: "Subsequently, we assessed T cell phenotype in the two groups of mice. While neither the frequencies nor absolute counts of

aortic CD4⁺ and CD8⁺ T cells differed significantly between two groups of mice (Figure 4D-F), CD69 frequency and CD44 MFI (Mean Fluorescence Intensity), the T cell activation markers, were significantly reduced in both CD4⁺ and CD8⁺ T cells from Xcr1⁺ cDC1 depleted mice compared to controls (Figure 4G and H)”.

(3) How does T cell activation mechanistically influence atherosclerosis progression? Why was CD69 selected as the sole activation marker? Were other markers (e.g., KLRG1, ICOS, CD44) examined to confirm activation status?

We sincerely appreciate these insightful comments. As extensively documented in the literature, activated effector T cells (both CD4⁺ and CD8⁺) critically promote plaque inflammation and instability through their production of pro-inflammatory cytokines (particularly IFN- γ and TNF- α), which drive endothelial activation, exacerbate macrophage inflammatory responses, and impair smooth muscle cell function [A].

In our study, we specifically investigated the role of cDC1 cells in atherosclerosis progression. Our key findings demonstrate that cDC1 depletion attenuates T cell activation (as shown by reduced CD69/CD44 expression) and that this reduction in activation is functionally linked to the observed decrease in atherosclerosis burden in our model.

Regarding CD44 as an activation marker, we performed quantitative analyses of CD44 mean fluorescence intensity (MFI) in aortic T cells (Figure 4). Importantly, the MFI of CD44 was significantly lower on both CD4⁺ and CD8⁺ T cells from Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} mice compared to the control ApoE^{-/-} mice (data shown below), which is consistent with the result of CD69 in Figure 4. We added the related description in the Result section.

Please refer to the Results section from line 185 to 187 “CD69 frequency and CD44 MFI (Mean Fluorescence Intensity), the T cell activation markers, were significantly reduced in both CD4⁺ and CD8⁺ T cells from Xcr1⁺ cDC1 depleted mice compared to controls (Figure 4G and H)”.

Similarly, MFI of CD44 was significantly lower on both CD4⁺ and CD8⁺ T cells from Xcl1^{-/-} ApoE^{-/-} mice compared to the control ApoE^{-/-} mice (data shown below), which is consistent with the result of CD69 in Figure 7. We also added the related description in the Result section.

Please refer to the Results section from line 308 to 309 “Crucially, CD69⁺ frequency and CD44 MFI remained comparable in both aortic CD4⁺ and CD8⁺ T cells between two groups (Figure 7D-F).”

[A] Hansson, Göran K, and Andreas Hermansson. “The immune system in atherosclerosis.” Nature immunology vol. 12,3 (2011): 204-12. doi:10.1038/ni.2001

(4) Figure 7B: Beyond cDC1/2 proportions within cDCs, please report absolute counts of: Total cDCs, cDC1, and cDC2 subsets. Figure 7D: In addition to CD4⁺/CD8⁺ T cell proportions, the following should be included:

(a) Total T cell numbers in aortas

(b) Absolute counts of CD4⁺ and CD8⁺ T cells.

Thanks for your suggestions. We have now included in Figure 7 the absolute counts of cDC, cDC1, and cDC2 cells, along with CD4⁺ and CD8⁺ T cells in aortic tissues. Additionally, we provide the corresponding CD44 mean fluorescence intensity (MFI) measurements for both CD4⁺ and CD8⁺ T cell populations. We added the related description in the Result section.

Please refer to the Results section from line 303 to 311: “The flow cytometric results illustrated that both frequencies and absolute counts of Xcr1⁺ cDC1 cells in the aorta were significantly reduced, but cDCs and cDC2 cells from Xcl1^{-/-} ApoE^{-/-} were comparable with that from ApoE^{-/-} (Figure 7A-C). Moreover, in both lymph node and spleen, the absolute numbers of pDC, cDC1 and cDC2 from Xcl1^{-/-} ApoE^{-/-} were comparable with that from ApoE^{-/-} (Figure S11). Crucially, CD69⁺ frequency and CD44 MFI remained comparable in both aortic CD4⁺ and CD8⁺ T cells between two groups (Figure 7D-F). However, aortic CD8⁺ T cells exhibited reduced frequency and absolute count, while CD4⁺ T cells showed increased frequency but unchanged counts in Xcl1^{-/-} ApoE^{-/-} mouse versus controls (Figure 7G and H).”

(5) cDC1 depletion reduced CD69+CD4+ and CD69+CD8+ T cells, whereas Xcl1 depletion decreased Xcr1+ cDC1 cells without altering activated T cells. How do the authors explain these different results? This discrepancy needs explanation.

We sincerely appreciate your professional and insightful comments regarding the mechanistic relationship between cDC1 depletion and T cell activation. Direct cDC1 depletion in the Xcr1^{Cre-Gfp} Rosa26^{LSL-DTA} ApoE^{-/-} micromodel removes both recruited and tissue-resident cDC1s, eliminating their multifunctional roles in antigen presentation, co-stimulation and cytokine secretion essential for T cell activation. In contrast, Xcl1 depletion reduces, but does not eliminate cDC1 migration into plaques. Furthermore, alternative chemokine axes (e.g., CCL5/CCR5, CXCL9/CXCR3, BCL9/BCL9L) may partially rescue cDC1 recruitment [13, 68, 69], and non-cDC1 APCs (e.g., monocytes, cDC2s) may compensate for T cell activation [55, 70]. We emphasize that Xcl1 depletion specifically failed to alter T cell activation in hyperlipidemic ApoE^{-/-} mice. However, its impact may differ in other pathophysiological contexts due to compensatory mechanisms. We thank you again for highlighting this nuance, which strengthens our mechanistic interpretation. We have added these points to the discussion section and included new references.

Please refer to the Discussion section from line 407 to 413: “Notably, while complete ablation of Xcr1⁺ cDC1s impaired T cell activation, reduction of Xcr1⁺ cDC1 recruitment via Xcl1 deletion did not significantly compromise this process. This discrepancy may arise through compensatory mechanisms: alternative chemokine axes (e.g., CCL5/CCR5, CXCL9/CXCR3, BCL9/BCL9L) may partially rescue Xcr1⁺ cDC1 homing [13, 68, 69], while non-cDC1 antigen-presenting cells (e.g., monocytes, cDC2s) may sustain T cell activation [55, 70]. Furthermore, tissue-specific microenvironment factors could potentially modulate its role in other diseases.”. [13] Eisenbarth, S C. “Dendritic cell subsets in T cell programming: location dictates function.” *Nature reviews. Immunology* vol. 19,2 (2019): 89-103. doi:10.1038/s41577-018-0088-1 [55] Brewitz, Anna et al. “CD8+ T Cells Orchestrate pDC-XCR1+ Dendritic Cell Spatial and Functional Cooperativity to Optimize Priming.” *Immunity* vol. 46,2 (2017): 205-219. doi:10.1016/j.immuni.2017.01.003 [68] de Oliveira, Carine Ervolino et al. “CCR5-Dependent Homing of T Regulatory Cells to the Tumor Microenvironment Contributes to Skin Squamous Cell Carcinoma Development.” *Molecular cancer therapeutics* vol. 16,12 (2017): 2871-2880. doi:10.1158/1535-7163.MCT-17-0341.[69] He F, Wu Z, Liu C, Zhu Y, Zhou Y, Tian E, et al. Targeting BCL9/BCL9L enhances antigen presentation by promoting conventional type 1 dendritic cell (cDC1) activation and tumor infiltration. *Signal Transduct Target Ther.* 2024;9(1):139. Epub 2024/05/30. doi: 10.1038/s41392-024-01838-9. PubMed PMID: 38811552; PubMed Central PMCID: PMCPMC11137111.[70] Böttcher, Jan P et al. “Functional classification of memory CD8(+) T cells by CX3CR1 expression.” *Nature communications* vol. 6 8306. 25 Sep. 2015, doi:10.1038/ncomms9306.

Reviewer #1 (Recommendations for the authors):

(1) Line 32 - The authors might want to add that the mouse model leads to a "constitutive" depletion of cDC1.

Thanks for your advice, we have revised the sentence as follows.

Please refer to the Results section from line 31 to 33: “we established $Xcr1^{Cre-Gfp}$ $Rosa26^{LSL-*DTA* ApoE^{-/-}}$ mice, a novel and complex genetic model, in which cDC1 was constitutively depleted in vivo during atherosclerosis development”.

(2) Line 187-188: The authors claim that T cell activation was "inhibited" if cDC1 was depleted. The data shows that the T cells were less activated, but there is no indication of any kind of inhibition; this should be corrected.

Thanks for your advice, we have revised the sentence as follows.

Please refer to the Results section from line 183 to 187: “Subsequently, we assessed T cell phenotype in the two groups of mice. While neither the frequencies nor absolute counts of aortic $CD4^{+}$ and $CD8^{+}$ T cells differed significantly between two groups of mice (Figure 4D-F), CD69 frequency and CD44 MFI (Mean Fluorescence Intensity), the T cell activation markers, were significantly reduced in both $CD4^{+}$ and $CD8^{+}$ T cells from $Xcr1^{+}$ cDC1 depleted mice compared to controls (Figure 4G and H)”.

(3) Why are some splenic DC clusters absent in LNs and vice versa? This is not obvious to this reviewer and should at least be discussed.

We appreciate the insightful question regarding the absence of certain splenic DC clusters in LNs. This phenomenon in Figure 5 aligns with the 'division of labor' paradigm in dendritic cell biology: tissue microenvironments evolve specialized DC subsets to address local immunological challenges. The absence of universal clusters reflects functional adaptation, not technical artifacts. We acknowledge that this tissue-specific heterogeneity warrants further discussion and have expanded our analysis to address this point in the discussion part of our manuscript.

Please refer to the Discussion section from line 375 to 385: “This pronounced tissue-specific compartmentalization of $Xcr1^{+}$ cDC1 subsets may related to multiple mechanisms including developmental imprinting that instructs precursor differentiation into transcriptionally distinct subpopulations [62], and microenvironmental filtering through organ-specific chemokine axes (e.g., CCL2/CCR2 in spleen) selectively recruits receptor-matched subsets [63, 64]. This spatial specialization optimizes pathogen surveillance for local immunological challenges. Based on the maturation analysis of the cDC1 scRNA seq data [41], our findings suggest that the aortic cDC1 cells display a major difference from those of spleen and lymph nodes by lacking the mature clusters, whereas lymph node cDC1 cells contain an additional $Fabp5^{+}$ $S100a4^{+}$ late mature Cluster. Our results also suggest that hyperlipidemia contributes to alteration in early immature cDC1 and in the abundance of late immature cDC1 cells, which was associated with dramatic change in gene expression of *Tnfrsf3*, *Serinc3*, *Apol7c* and *Tifab*”.

[62]. Liu Z, Gu Y, Chakarov S, Bleriot C, Kwok I, Chen X, et al. Fate Mapping via Ms4a3-Expression History Traces Monocyte-Derived Cells. *Cell*. 2019;178(6):1509-25 e19. Epub 2019/09/07. doi: 10.1016/j.cell.2019.08.009. PubMed PMID: 31491389.

[63]. Bosmans LA, van Tiel CM, Aarts S, Willemsen L, Baardman J, van Os BW, et al. Myeloid CD40 deficiency reduces atherosclerosis by impairing macrophages' transition into a pro-inflammatory state. *Cardiovasc Res*. 2023;119(5):1146-60. Epub 2022/05/20. doi: 10.1093/cvr/cvac084. PubMed PMID: 35587037; PubMed Central PMCID: PMCPCMC10202633.

[64]. Mildner A, Schonheit J, Giladi A, David E, Lara-Astiaso D, Lorenzo-Vivas E, et al. Genomic Characterization of Murine Monocytes Reveals C/EBPbeta Transcription Factor Dependence

of Ly6C(-) Cells. *Immunity*. 2017;46(5):849-62 e7. Epub 2017/05/18. doi: 10.1016/j.immuni.2017.04.018. PubMed PMID: 28514690.

[41]. Bosteels V, Marechal S, De Nolf C, Rennen S, Maelfait J, Tavernier SJ, et al. LXR signaling controls homeostatic dendritic cell maturation. *Sci Immunol*. 2023;8(83):eadd3955. Epub 2023/05/12. doi: 10.1126/sciimmunol.add3955. PubMed PMID: 37172103.

(4) The authors should discuss how XCL1 could impact lesional cDC1 and T cell abundance. Notably, preDCs do not express XCR1, and T cells express XCL1 following TCR activation. Is there a recruitment or local proliferation defect of cDC1 in the absence of XCL1? Could there also be a role for NK cells as a potential source of XCL1?

We appreciate your insightful questions regarding the differential effects of Xcl1 on cDC1s and T cells. Xcl1 primarily mediates the recruitment of mature cDC1s. Our data demonstrate that Xcl1 deletion significantly reduces aortic cDC1 abundance, which correlates with a concomitant decrease in CD8⁺ T cell numbers within the aorta. These findings strongly suggest that the Xcl1-Xcr1 axis plays a regulatory role in T cell accumulation in aortic plaques.

Consistent with prior studies [A, B], cDC1 recruitment can occur in the absence of Xcl1 which echoes our findings that cDC1 cells were still found in Xcl1 knockout aortic plaque but in lower abundance. It is very true that further studies are required to address how the Xcl1 dependent and independent cDC1 cells activate T cells and if they possess capability of proliferation in tissue differentially. We have added these points in discussion section.

Please refer to the Discussion section from line 407 to 415: “Notably, while complete ablation of Xcr1⁺ cDC1s impaired T cell activation, reduction of Xcr1⁺ cDC1 recruitment via Xcl1 deletion did not significantly compromise this process. This discrepancy may arise through compensatory mechanisms: alternative chemokine axes (e.g., CCL5/CCR5, CXCL9/CXCR3, BCL9/BCL9L) may partially rescue Xcr1⁺ cDC1 homing [13, 68, 69], while non-cDC1 antigen-presenting cells (e.g., monocytes, cDC2s) may sustain T cell activation [55, 70]. Furthermore, tissue-specific microenvironment factors could potentially modulate its role in other diseases. In summary, our findings identify Xcl1 as a potential therapeutic target for atherosclerosis therapy, though its cellular origins and regulation of lesional Xcr1⁺ cDC1 and T cells dynamics require further studies”.

In literatures, Xcl1 are expressed in NK cells and subsets of T cells, and NK cells can be a potential source of Xcl1 during atherosclerosis which deserve further investigations [A, C, D].

[A] Böttcher, Jan P et al. “NK Cells Stimulate Recruitment of cDC1 into the Tumor Microenvironment Promoting Cancer Immune Control.” *Cell* vol. 172,5 (2018): 1022-1037.e14. doi:10.1016/j.cell.2018.01.004

[B] He, Fenglian et al. “Targeting BCL9/BCL9L enhances antigen presentation by promoting conventional type 1 dendritic cell (cDC1) activation and tumor infiltration.” *Signal transduction and targeted therapy* vol. 9,1 139. 29 May. 2024, doi:10.1038/s41392-024-01838-9

[C] Woo, Yeon Duk et al. “The invariant natural killer T cell-mediated chemokine X-C motif chemokine ligand 1-X-C motif chemokine receptor 1 axis promotes allergic airway hyperresponsiveness by recruiting CD103⁺ dendritic cells.” *The Journal of allergy and clinical immunology* vol. 142,6 (2018): 1781-1792.e12. doi:10.1016/j.jaci.2017.12.1005

[D] Winkels, Holger et al. “Atlas of the Immune Cell Repertoire in Mouse Atherosclerosis Defined by Single-Cell RNA-Sequencing and Mass Cytometry.” *Circulation research* vol. 122,12 (2018): 1675-1688. doi:10.1161/CIRCRESAHA.117.312513

Reviewer #2 (Recommendations for the authors):

There is a logical error in line 298. I suggest revising to: "Collectively, these data suggest that Xcl1 promotes atherosclerosis by recruiting Xcr1+ cDC1 cells, which subsequently drive T cell activation in lesions."

Thanks for your advice. Since Xcl1 deficiency reduced both the frequencies and absolute counts of Xcr1+ cDC1 and CD8+ T cells in lesions without affecting T cell activation, we revised the sentence as you suggested.

Please refer to the Results section from line 314 to 315: "Collectively, these data suggest that Xcl1 promotes atherosclerosis by recruiting Xcr1⁺ cDC1 cells, and facilitating CD8⁺ T cell accumulation in lesions".

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